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Inductive Noise Spectroscopy of Ferromagnetic Thin-Films

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Abstract

This work explores the use of inductive magnon noise spectroscopy (iMNS) as a passive characterisation technique for equilibrium magnetisation dynamics in ferro/ferrimagnetic thin films, building upon recent work in detection of thermally excited magnons by Siegl et al. [1]. Two coplanar waveguides (CPWs) were designed; simulated using COMSOL; and manufactured, with the U-shaped design selected for its superior transmission characteristics. Broadband vector network analyser ferromagnetic resonance (VNA-FMR) experiments were performed on yttrium iron garnet (YIG) and Permalloy (Py) thin films, yielding Gilbert damping parameters of $\alpha = 0.0021 \pm 0.0008$ and $\alpha = 0.009 \pm 0.002$ respectively, consistent with literature values. MuMax3 micromagnetic simulations were carried out to provide theoretical benchmarks, showing good agreement with experimental results. iMNS simulations confirmed the feasibility of detecting resonant contrast at the Kittel frequency for both samples. However, experimental iMNS results were inconclusive, with anomalous noise artefacts attributed to waveguide damage, electromagnet hysteresis, and spectrum analyser limitations obscuring the expected resonant contrast. The prospect of replacing the cryogenic termination with a maser mode cooled signal source is discussed as a route towards room temperature iMNS without the need for cryogenic cooling.

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AI declaration

Artificial intelligence (Ai), including GenAi has not been used in writing any part of this project. The only exception was in the occasional aid in writing code used for data processing wherein OpenAI ChatGPT-5 (https://chatgpt.com/) was used.

Collaboration and supervision

The work carried out in this report is completely independent from any other work carried out beforehand. Some code used in this project to calibrate the electromagnet was provided to me by Adam Parsons, a PhD candidate in the research group and was subsequently modified for the use-case.

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Chapter 1

Aims and Context

1.1 Spintronics and Magnonics

Spintronics is the field of electronics which utilises the spin degree of freedom in charge carriers. It aims to manipulate spin currents and their interactions for the development of next generation devices.[2] This has created large advancements in data storage, developing non-volatility, low power dissipation, and high processing rates.[3,4] However, traditional spintronic technologies -such as a spin polarised currents- depend on the movement of electrons for information transfer and processing, where there is a component of Joule heating dissipation that cannot be avoided.[5] Magnon spintronics or magnonics is an increasingly important subfield of spintronics which utilises spin currents carried by magnons -the quanta of spin waves- for information processing, sensing and other devices.[6] Magnonics has two main advantages over traditional spintronics:

- Magnons can propagate centimetre distances free from Joule heating dissipation, as there is no carrier movement only spin.[5,6]
- Magnons can propagate in any form of magnetically ordered materials, including insulating systems -which prevent dissipation due to eddy currents, and antiferromagnets -preventing stray magnetic fields.[7]

The development of magnonic devices has led to increasing interest in understanding the magnetisation dynamics of ferromagnetic materials. Existing magnetic devices typically operate in the microwave regime, between 300 MHz - 300 GHz, however antiferromagnetic materials could be operated up to terahertz frequencies. As such, the need to improve detection technology for these mechanisms is paramount.[4]
In the case of magnon currents, there are two main kinds: coherent (ballistic), and incoherent (diffusive) currents. Coherent currents have a well defined frequency, wavelength, and phase. For sufficiently high coherency, information can be encoded into the amplitude and phase of the spin wave, which enables for the interference-based, wave processing devices.[4,8]
In addition to coherency based interference, magnons strongly interact with microwave photons, which allows for classical and quantum processing, but may suffer from thermal photon interference. This has a critical functionality in the implementation to existing optical fibre communication.[8,9]

Magnetisation Dynamics

The mechanism behind a ferromagnet's magnetisation dynamics can be described by the precessional motion of magnetic moments due to the influence of an effective magnetic field. A single electron spin has a magnetic dipole moment, $\boldsymbol{\mu}$ which is proportional to its angular momentum $\boldsymbol{J}$ by:

$$\boldsymbol{\mu} = \gamma \boldsymbol{J} \tag{1.1}$$

Where $\gamma = g\mu_B/\hbar$ is the gyromagnetic ratio, g is the g-factor of the electron, μ_B is the Bohr magneton, and $\hbar$ is the reduced Planck constant. A free electron has a g-factor, $g \approx 2$ resulting in a reduced gyromagnetic ratio, $\gamma/2\pi \approx 28\,\mathrm{GHz/T}$, as such simple laboratory equipment which can generate fields in the order of $1 \times 10^{-1}\,\mathrm{T}$, we expect to observe resonance with perturbing radiation in the $\approx 3\,\mathrm{GHz}$ range.

When the electron is in the presence of an external (effective) magnetic field $\boldsymbol{B}_{\text{eff}}$, a torque is exerted on the magnetic moment which will tend to line up parallel to the field in order to minimise the magnetostatic energy.

$$\boldsymbol{\tau} = \boldsymbol{\mu} \times \boldsymbol{B}_{\text{eff}} = \gamma \boldsymbol{J} \times \boldsymbol{B}_{\text{eff}} \tag{1.2}$$

From Newton's second law for rotation, we can use the general relation $\boldsymbol{\tau} = \mathrm{d}\boldsymbol{J}/\mathrm{d}t$ such that the equation of motion for $\boldsymbol{J}$ can be written as:

$$\frac{\mathrm{d}\boldsymbol{J}}{\mathrm{d}t} = \boldsymbol{\mu} \times \boldsymbol{B}_{\text{eff}} \tag{1.3}$$

Where $\boldsymbol{B}_{\text{eff}} = \mu_0 \boldsymbol{H}_{\text{eff}}$, μ_0 is the vacuum permeability, and $\boldsymbol{H}_{\text{eff}}$ is the effective magnetic field. The effective magnetic field is the sum of the externally applied static field $\boldsymbol{H}_0$, the field from exchange interactions, $\boldsymbol{H}_{\text{ex}}$, the demagnetising field, $\boldsymbol{H}_{\text{demag}}$, and fields due to anisotropies, $\boldsymbol{H}_{\text{ani}}$, as well as the dynamic component of the externally applied magnetic field, $\boldsymbol{h}_{\text{appl}}(t)$.
By treating the individual spins as one large macro-spin, the bulk magnetisation, $\boldsymbol{M}$ starts to precess at the Larmor frequency, $\omega = \gamma \mu_0 \boldsymbol{H}_{\text{eff}}$. Which is described by the Landau-Lifshitz (LL) equation:

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = -\gamma \mu_0 \boldsymbol{M} \times \boldsymbol{H}_{\text{eff}} \qquad \text{(Landau-Lifshitz Equation)} \tag{1.4}$$

The LL equation fails to describe real materials' behaviour, as according to the equation, the magnetisation, $\boldsymbol{M}$ does not align to $\boldsymbol{H}_0$. This relaxation originates in electron interactions with phonons, magnons, and lattice defects and is described phenomenologically by the Landau-Lifshitz-Gilbert[10] (LLG) equation which includes a viscous damping term:

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = \underbrace{-\gamma \mu_0 \boldsymbol{M} \times \boldsymbol{H}_{\text{eff}}}_{\text{precession term}} + \underbrace{\frac{1}{M_s} \boldsymbol{M} \times \alpha \frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t}}_{\text{damping term}} \qquad \text{(Landau-Lifshitz-Gilbert Equation)} \tag{1.5}$$

Where α is the dimensionless Gilbert damping parameter. Ample research has been done showing this parameter changes with sample geometry, spin pumping, and other magnon interactions (two-magnon due to surface roughness, magnon-phonon, and higher-order magnon-magnon scattering).[5,11–14] Consequently, the spin behaviour during equilibration traces a helical path (Figure 2.1a) which aligns to $\boldsymbol{H}_0$.

Material Requirements

Magnon band structures can be engineered to great effect, as they are highly dependent on various energy contributions from sample geometry and the magnetic properties of the materials.[8] From a material science perspective, the parameters that can be varied to tune the spectrum include the saturation magnetisation, M_s, the film thickness, d, and the exchange stiffness, η.
For the purpose of high frequency devices such as memory devices, a high magnetic anisotropy is necessary to stabilise the magnetic domains. The Gilbert damping of a material describes the relaxation of a magnetic moment towards equilibrium, and can thus be used to infer how quickly a magnetic moment aligns to an applied field.
Suitable materials for high frequency spintronic devices have a low Gilbert damping parameter, characterised by a narrow ferro- or ferrimagnetic linewidth in experiments.[15,16] Consequently, for the development of spintronic and magnonic devices, materials with a narrower linewidth, corresponding to a low Gilbert damping are necessary, alongside a simple, and reliable way to characterise.

Existing Experimental Methods

- Pulsed Inductive Microwave Magnetometer (PIMM): PIMM is a robust and flexible measurement technique that permits rapid evaluation of thin-film materials for data storage and rf applications, it can measure FMR frequencies, high frequency damping, and dynamical anisotropy.[17]
- Brillouin Light Scattering (BLS): BLS is a method used to study low energy magnons, phonons, and their interactions. Useful to measure anisotropy fields and exchange interactions.[18]
- Magnetisation State Tomography (MST): MST is a technique to build a representation of a quantum system from a set of measurements of a system's quadratures -operators which represent the complex

and real and imaginary components of the complex amplitude of the complex amplitude of the system's creation and annihilation operators-[19] taken from different forms of measurements but most accurately, the inverse spin hall effect (ISHE) which is the conversion of a charge current into a transverse spin current via spin-orbit coupling.[20]

- Ferromagnetic resonance (FMR): FMR is a spectroscopic technique which looks at the magnetisation dynamics and relaxation of ferro and ferrimagnetic materials. It is simple and widely used and can be used to a high degree of accuracy.
- Time Resolved Magneto-Optic Kerr Effect (MOKE)
- Inductive Magnon Noise Spectroscopy (iMNS): iMNS is a conceptually simple, broadband technique for the characterisation of thermal fluctuations in magnetic samples. It is a type of state tomography for the quantum state of a system allowing for detection of non equilibrium magnonic states. It operates on the principle that if a signal is out of thermal equilibrium with the magnetic sample, there will be a net energy flux only at the resonance frequency. As such the magnetisation dynamics can be measured passively when the signal source is the intrinsic emission of electrical components.[1]
  Importantly, this is a passive characterisation technique, which relies only on thermal gradients which do not perturb the magnetic states.

Yttrium Iron Garnet (YIG)

YIG has been at the centre of spintronic research for the better part of the 21st century[15,21] for its many attractive properties in the manufacturing and operation of future devices. It lies on the junction between traditional electronics, spintronics, and magnonics, as an ideal material to convert magnon excitations into spin and charge currents through ISHE and spin pumping mechanisms.[20,21]
YIG, $Y_3Fe_5O_{12}$ is a magnetic garnet which are insulating, ionic solids with a general formula $R_3{-}Fe_2(FeO_4)_3$ in which the five iron(III) ions occupy three tetrahedral, and two octahedral sites, and three dodecahedral sites occupied by the (in this case) Yttrium ions. The exchange interactions are such that the magnetic moments of the tetrahedral cations tend to be antiparallel to both the octahedral and dodecahedral cations.[15]
YIG has a cubic magneto-crystalline anisotropy with an easy axis along the (111) direction. It is a ferrimagnet known for remarkably narrow linewidths and low damping which can be attributed to its ferric ions, which contribute to the magnetic moments, being in an $L = 0$ state (no orbital moments), leading to weak lattice interaction when epitaxially grown.[5,8,21] Further, as an insulator the reduced conductivity in YIG results in reduced eddy currents, which often dominate damping and allow for smaller α values.
In the context of devices, YIG has the disadvantage of a large unit cell of 1.2376 nm[8] which limits the minimum thickness of a YIG film to ≈ 5 nm, increasing complexity to integrating with existing semiconductor technologies.

Permalloy (Py)

Fe−Ni alloys are crucial soft magnetic alloys, Ni rich alloys are called permalloys (Py), with a typical composition of $Ni_{80}Fe_{20}$. Small variations in the composition leads to different properties which can be exploited for different applications such as high permeability, small coercivity, and near zero magnetostriction. These different compositions include alloys such as supermalloy, or Mu-metal.
As previously discussed, materials with low Gilbert damping are required for spintronic device operation. Permalloy has been researched and used for race track memory, magnetic sensors, and magnetic shielding.
It is generally accepted in literature for the saturation magnetisation of Py to be $M_S^{\text{lit}} = 8.6 \times 10^5 \text{A m}^{-1}$, which is often used in micromagnetic simulations and analytical calculations. The Gilbert damping parameters are generally accepted to range between 0.007 to 0.05. It has been observed that annealing, high temperatures, and sample thickness all influence this value.[3,22,23]

1.2 Research Focus

The aim of this project is to replicate and expand upon the work carried out by Siegl et al. [1]. Their work explores the possibility of using inductive noise spectroscopy (iMNS) for the detection of resonant magnetic properties at equilibrium for a yttrium-iron-garnet (YIG) thin film.

To replicate their experiments the project involves the design of a coplanar waveguide (CPW) used to discern the microwave emissions by the samples at room temperature; vector network analyser (broadband) ferromagnetic resonance (VNA-FMR) experiments utilising this waveguide; and iMNS to observe equilibrium magnon mode frequencies of the samples. In this work, YIG and Permalloy films are used which are further analysed using superconducting quantum interference device (SQUID) to verify experimental results.
The work aims to explore the horizon of iMNS as a characterisation technique. A desirable outcome would be to replace the actively cooled 50 Ohm noise source with room temperature sources of 'cold' signals and 'hot' noise. This includes the possibility of incorporating a maser mode cooled signal as the low temperature noise source to expand upon the work of Siegl et al. [1] and discuss the practical implications of its implementation for future experiments without the need for cryogenic cooling.

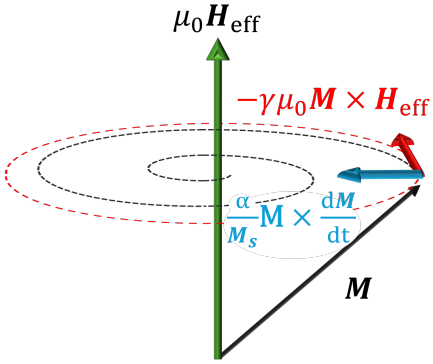
Chapter 2

Background

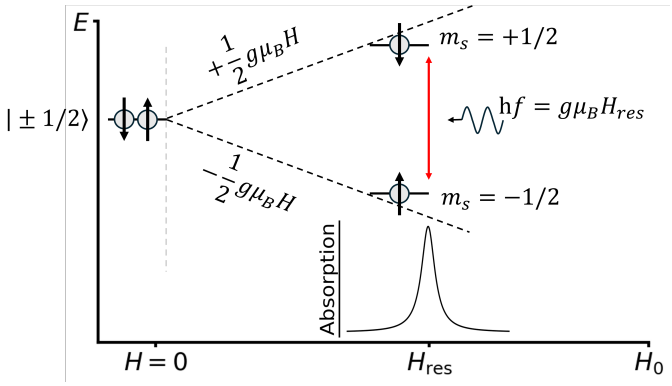
2.1 Ferromagnetic Resonance (FMR)

Resonance Conditions

Equation 1.5 can be used to describe FMR based on two different magnetic fields. The applied static field $\boldsymbol{H}_0$ which saturates the magnetic moment in the direction of the field, and a pumping MW field which excites the total magnetic moment of the sample to precess around the direction of $\boldsymbol{H}_0$ at the Larmor frequency. Microscopically, the $\boldsymbol{H}$ field creates a Zeeman splitting of the energy levels, the MW excites the magnetic dipole transitions between these split levels. The energy of the transverse MW field is most strongly absorbed when its frequency coincides with the Larmor frequency[24] (Figure 2.1b) and can be measured using a vector network analyser (VNA).



(a) Magnetisation precession draws out a helical path towards applied field. Precession (red) at Larmor frequency and the damping (blue). MW field is absorbed when MW frequency coincides with Larmor frequency.



(b) Energy diagram for a spin half system. The $\boldsymbol{H}$ field creates Zeeman splitting wherein MW radiation is absorbed at frequency corresponding to the energy gap.

Equation 1.5 can be analytically solved by decomposing $\boldsymbol{M}$ and $\boldsymbol{H}$ into their static and dynamic terms, and assuming that the dynamic terms are much smaller than their static counterparts, thus obtaining what is known as the Kittel equation:

$$\begin{aligned} f_{\text{res}} &= \frac{\gamma\mu_0}{2\pi}\sqrt{\boldsymbol{H}_1 \cdot \boldsymbol{H}_2} \\ &= \gamma_0'\sqrt{(\boldsymbol{H} + (N_{yy} - N_{zz})M_z) \cdot (\boldsymbol{H} + (N_{xx} - N_{zz})M_z)} \end{aligned} \tag{2.1}$$

Where $\gamma_0' = \frac{\gamma\mu_0}{2\pi}$ is the reduced gyromagnetic ratio, $\boldsymbol{H}_1$ and $\boldsymbol{H}_2$ are the corresponding static and varying fields characterising the torque exerted on the magnetisation and N_{ij} are the components of the demagnetising tensor:

$$N = \begin{pmatrix} N_{xx} & 0 & 0 \\ 0 & N_{yy} & 0 \\ 0 & 0 & N_{zz} \end{pmatrix} \tag{2.2}$$

for which its trace ($N_{xx} + N_{yy} + N_{zz}$) is equal to 1. A more generalised expression for the FMR frequency was derived by J. Smit and H. G. Beljers [25] which uses an energy landscape to account for orientational perturbation and crystal anisotropies. Its benefits are apparent in larger samples, low frequency experiments, and in samples with unknown crystal orientations.[26–28] As such for the scope of this research the Kittel equation (2.1) is utilised for these experiments.

Thin Films

In a thin-film, the demagnetisation fields associated with the surface charges dominate, corresponding to all the components in the demagnetisation tensor in equation 2.2 being 0 except for $N_{zz} = 1$, leading to an elliptically precessing magnetisation direction allowing for a range of complex dipolar and exchange interactions. (see section 2.2.)[8]
By applying thin-film constraints to equation 2.1 and more specifically, to the demagnetisation tensor, the in-plane and out-of-plane resonance conditions[1] become:

$$\begin{aligned} \text{In-plane: } f_{\text{res}} &= \gamma_0' \sqrt{\boldsymbol{H}(\boldsymbol{H} + \boldsymbol{M}_{\text{eff}})} \\ \text{Out of plane: } f_{\text{res}} &= \gamma_0' (\boldsymbol{H} - \boldsymbol{M}_{\text{eff}}) \end{aligned} \tag{2.3}$$

This doesn't take into account that any field in the same form as the demagnetising field will be indistinguishable from it. As a result, the effective magnetisation includes anisotropy fields, and has been reported to different degrees of precision depending on the specific experiment conditions.[21,29]
Typically in thin-films the assumption is made that all the magnetic moments across the film thickness are parallel because of the strong exchange interactions between spins. As such the magnetic moment of the film can be approximated to the algebraic sum of all its constituent moments, assuming ideally smooth interfaces.

Linewidth

Key information is obtained from the resulting lineshape due to it being proportional to the imaginary part of the complex susceptibility (appendix D), which rarely deviates from a standard Lorentzian lineshape.[20]
When performing a frequency swept experiment, a further parameter is then obtained as the frequency full width at half maximum, Δf_{FWHM}, which describes the intrinsic damping contributions. If the Gilbert damping represents the entire intrinsic damping, then the linewidth will depend linearly on the perturbation frequency.[12]

$$\Delta f_{\text{FWHM}} = 2\alpha f \tag{2.4}$$

From a macroscopic point of view, the linewidth describes the viscous damping of the magnetisation motion.[24]
It is worth highlighting that linear dependence between linewidth and frequency is only when the magnetisation and external magnetic field are parallel. Otherwise the field-dragging effects need to be included.[12] This would expand the equation further for inhomogeneous broadening effects.
Field linewidth is typically reported when performing field swept experiments, and is typically represented as the peak-to-peak of the inflection points of the imaginary susceptibility. In most cases the relationship between the two linewidths can be estimated using:[30,31]

$$\Delta H = \alpha \left(\frac{4\pi f}{\gamma} \right) + \Delta H_0 \tag{2.5}$$

The linewidth ΔH is directly linked to the relaxation process. Other than the Gilbert damping, other relaxation paths include spin-pumping effects, and magnon scattering mechanisms which depend on sample thickness, crystallographic faults, etc. The inhomogeneous broadening, (ΔH_0) contribution is not dependent on frequency or orientation and arises from magnetic inhomogeneities.[12,30]
It has been widely observed that the Gilbert parameter α is frequently larger in thin films compared to bulk materials.[13,32] As previously discussed, the geometry of thin and ultra-thin films constraints the magnetic moments, as such surface roughness has a greater contribution towards the magnetisation dynamics in thin films. More importantly, the extrinsic, surface defects play a role which is indiscernible from intrinsic and defect induced effects.[32]

[1] Note that the resonant frequency is often denoted by the angular frequency $\omega_{\text{res}} = 2\pi f_{\text{res}}$ or as $\frac{\omega}{\gamma}$.

2.2 Spin Waves

The LLG equation describes the collective precessional motion of magnetic moments phenomenologically. This motion of magnetic moments is called a spin wave (SW). The magnetisation of a magnetically ordered material is mainly determined by the electron spin magnetic moment, $\mu_e^s = \pm\gamma\hbar/2$. Consequently, the quanta of a SW -called magnons- are bosonic quasiparticles associated with the flipping of one electron spin. As a result of being bosonic, magnons are described by Bose-Einstein statistics, and the intensity of a SW is characterised by its occupation number.[8]

Spin Waves in FMR

In terms of spin waves, FMR is the spatially uniform precession of the magnetisation across the whole sample. In other words, it corresponds to the lowest energy magnon mode with wavevector $k = 0$, commonly referred to as the "Kittel mode" in thin films.[24]

Magnon interactions play a key role in the magnetisation damping, particularly when in a multilayer array. It is worth noting, the magnon mechanisms described here are only operative when the magnetisation is in plane. For they rely on short-wavelength SWs which are not seen when magnetisation is normal to the film surface.[32]

Two-magnon scattering is a type of magnon interaction which is known to contribute to linewidth broadening in FMR. Wherein the excited $k = 0$ magnon scatters into degenerate magnon states with the same frequency but different wavevector $k \neq 0$.[12] This interaction is prevalent even in high quality single crystal YIG.[8]

In order to identify and characterise magnonic states, a baseline in thermal equilibrium must be identified. Magnons have frequencies which lie in the GHz regime, as such they are particularly attractive for future quantum technologies. This also means they are hard to detect as their thermal fluctuations are hidden by thermal J-N noise. (See section 2.4)

2.3 Coplanar Waveguide (CPW) Design

Coplanar waveguides (CPWs) are commonplace in MW electronics due to their low losses, and broadband support. They have become standard in FMR experiments due to the fact samples do not need to be remounted and new cavities don't need to be manufactured.[13] They consist of a centre strip conductor separated from semi-infinite ground plates on either side by a gap. This is printed onto a dielectric substrate in which the magnetic field is primarily focused -leading to lower losses. A grounded coplanar waveguide (G-CPW) refers to a CPW with a ground plane below the dielectric substrate which acts as a shield, containing the magnetic flux to the substrate.[33]

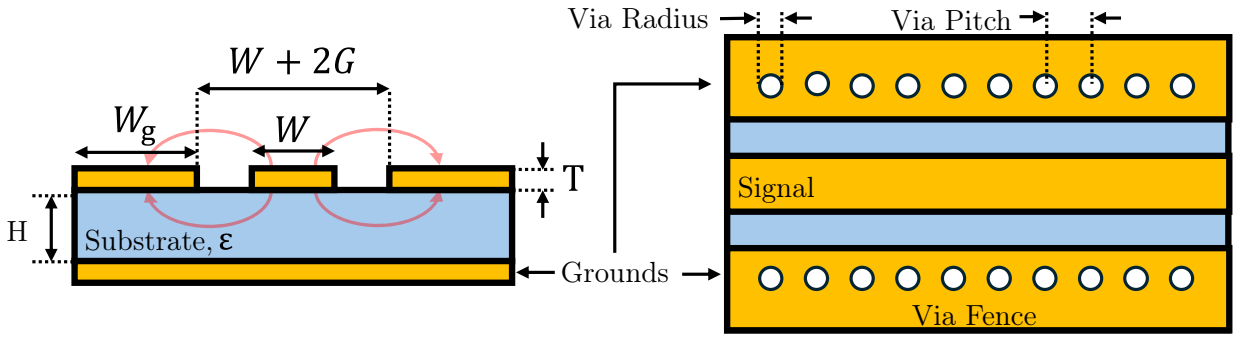


Figure 2.2: Simplified schematic of CPW geometry from side and plan view. Red arrows depict quasi-TEM mode.

The dimensions of each of the main components are chosen as to ensure quasi-TEM (transverse electromagnetic) mode of propagation. As a result the design must follow the following constraints (Appendix A):

- The width of the ground planes should be wider than the gap between them, $W_g > W + 2G$
- The spacing between the ground planes should be smaller than a half-wavelength of signal, $W + 2G \ll \lambda/2$[34]

When considering the application of a CPW in experiment, one must consider the sample does not short the CPW itself. As such an insulating layer or photoresist can be coated onto the waveguide or sample. However, this could add a layer of noise to a signal. As such kapton tape has been typically used to be placed between the waveguide and the sample.[35] It is worth noting that in some FMR experiments the surface magnon interactions can interact with the kapton tape and appear as a peak.[36] This should not play a role for YIG samples, due to its low conductivity.

In terms of manufacturing, it is necessary to avoid using a nickel coating or another ferromagnetic material as a coating or adhesive layer. Oliveira Schneider et al. [37] recommend using a tin surface coating without a nickel adhesive layer.

Characteristic Impedance

In microwave engineering, impedance matching is a key concept. The reasoning for this comes from the behaviour when the impedance along a transmission line changes, which results in reflections along the line. It is described by the scattering matrix. Which for an ideal lossless transmission line with impedance Z_0 that is terminated by a load with impedance Z_1 is given by:

$$|\Gamma| = \left| \frac{Z_1 - Z_0}{Z_1 + Z_0} \right| \tag{2.6}$$

As such when connecting the waveguide to a larger system, we must ensure it is impedance matched to prevent unnecessary radiative losses.

The transmission and reflections are measured by a vector network analyser (VNA) set up, which is widely used for FMR. They can be simply worked out using the scattering matrix:

$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \tag{2.7}$$

Where b_1 and b_2 are the sum of outgoing and incident waves at port 1 and 2 respectively. As such the power loss can be calculated from

$$\text{Power loss} = 1 - |S_{11}|^2 - |S_{21}|^2 \tag{2.8}$$

The impedance of a CPW can be calculated based on many properties of the materials used and geometry and is described in appendix A. The impedance is characteristic to the centre strip width, W and the width of the slots G. There are many useful resources to estimate these calculations[38] which coincide with the calculations described by Rainee N. Simons [39]

2.4 Inductive Magnon Noise Spectroscopy (iMNS)

Signal Noise

Electronic signals have a noise floor level which is set by thermal fluctuations of the electrons. This noise is called the Johnson-Nyquist (J-N) noise and at sub-terahertz frequencies[40] is described by its spectral density:

$$S_{\mathrm{n}} = 4k_{\mathrm{B}}TR \tag{2.9}$$

Importantly, J-N noise is proportional to the resistance and the temperature of a resistive component. As such any signal noise can also be described by its 'effective temperature'.[41,42] When measuring signals, particularly narrowband signals, shot-noise -noise due to random tunnelling events by the electrons- becomes relevant. However, unlike J-N noise, shot-noise is independent of temperature and resistance, and is only present when current is flowing.[43]

In an experimental set-up, a signal noise is typically measured using a spectrum analyser (SA) which functions by averaging the power within a frequency (radio) bandwidth, RBW. The RBW used experimentally, must be narrow enough to capture the signal linewidth, while remaining broad enough to increase the power measured, reduce measuring times, and mitigate shot-noise spikes.

Theory

A MW circuit in the presence of a thermal gradient will see a flow of energy along the transmission lines in the form of an imbalance of electromagnetic radiated noise in the MW frequencies. Consequently, a circuit which is in thermal equilibrium will see no net energy flux.
Importantly however, if a component of this circuit is outside of thermal equilibrium with the rest of the circuit, its MW emission changes proportional to its temperature. If this is the case energy will be drawn from the rest of the circuit. When a magnetic field is applied, a ferromagnetic sample will precess as discussed in section 2.1. This magnetisation precession most strongly exchanges energy at its resonance frequency. As such the emission induced by thermal fluctuations results in a net energy flow into the circuit with a non-uniform distribution -wherein we observe an enhanced detection of MW power at the Kittel frequency corresponding to standard FMR experiments- which can be discerned from the background noise.

Experimental Set-up

Using a broadband CPW to measure the emitted MW fluctuations of the thin film sample at room temperature, while connecting one end of the waveguide to a termination in a variable temperature environment, one can discern the noise due to the magnetisation precession. In this set up, the temperature of the magnetic system, and the MW system can be controlled independently of the underlying lattice by controlling the temperature of the termination.
In the set up used by Siegl et al. [1], a passive $50\,\Omega$ termination serves as a remote heat bath which is coupled to the magnetic system via the waveguide. As such when the magnetic system and the cold MW system are not in thermal equilibrium, a net flux of energy is detected which couples most strongly at the Kittel frequency.

2.5 Maser Mode Cooling

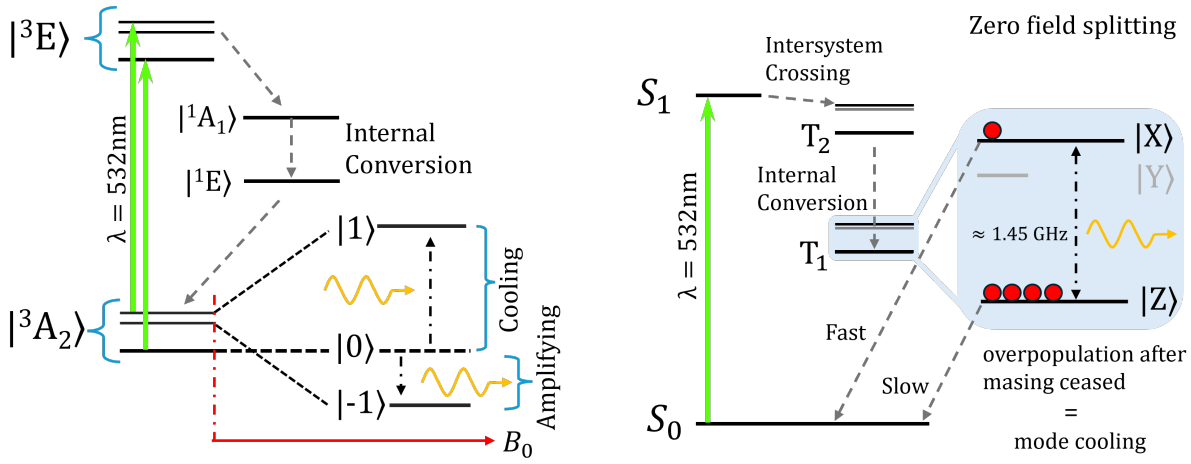
In section 2.4 it was established that the experiment relies on a cryogenically cooled load. This is typically a resistor termination cooled in liquid nitrogen to lower the noise temperature associated with J-N noise proportional to the temperature and the resistance. J-N sets the noise floor for MW measurements.
Masers are established devices used for signal amplification and are well known to be able to create "spin cold" MW signals.

Theory

Masing occurs from creating a population inversion between two energy levels separated by a MW frequency. As such its stimulated emission is in the corresponding MW regime. However, these devices can be operated in a stimulated absorption set up, in which the maser absorbs thermal photons in the linewidth of the corresponding quantum transition.[44]
Two masing systems which are often discussed are nitrogen vacancy (NV-) diamond, and pentacene. For both of which are placed within a resonator cavity to uniformly drive a spin ensemble. NV- diamond is an 8-level system which can focus on the ground state triplet where the masing action occurs as a population inversion across the $m_s = \pm 1$ - $m_s = 0$ transitions when a D.C. magnetic field is used to Zeeman split these levels (Figure 2.3b). In regular masing the $|0\rangle \leftrightarrow |-1\rangle$ transition is tuned to the resonance of the cavity, and the Zeeman splitting is such that the $|-1\rangle$ state is lower than the $|0\rangle$. For cooling, the $|0\rangle \leftrightarrow |+1\rangle$ transition is used.[45]
Pentacene is an 8 level system which can be treated as a 5 level system due to the spin preserving $|T_2\rangle \rightarrow |T_1\rangle$ transition (Figure 2.3a). Masing occurs in the $|X\rangle \leftrightarrow |Z\rangle$ transition in which the rate of the $|Z\rangle \leftrightarrow |S_0\rangle$ creates cooling when the $|X\rangle$ state is populated more slowly.[46,47]
If we think of a specific mode in a resonator, and assume all the noise is from black body radiation, then the average number of photons at the resonant frequency in equilibrium is given by:

$$\langle n \rangle = \frac{1}{\exp\left(\frac{hf}{k_B T}\right) - 1} \approx \frac{k_\mathrm{B} T}{hf} \tag{2.10}$$

Where the approximation is valid when $hf \ll k_B T$. If there is a spin system interacting with these photons, the two systems will try to reach equilibrium. The effective spin temperature can be calculated from the



(a) Jablonski diagram of NV- diamond. Zeeman splitting is shown for masing.

(b) Jablonski diagram of pentacene. Zero field splitting is depicted.

Figure 2.3: Jablonski diagrams for the masing systems discussed. Highlighting the pathway for mode cooling

population at the different energy levels:

$$\frac{1}{T} = \frac{k_B}{\Delta E} \ln\left(\frac{N - N_e}{N_e}\right) \tag{2.11}$$

Where N is the ensemble of spins which can exist in one of two states. N_e is the number of excited spins, and ΔE is the difference in energy between the two levels.
Then in increasing temperature: we see that when all states are at ground state, $T = 0K$; when there is some excited states, $T > 0K$; when half of all states are excited, $T = +\infty$; when there are more excited states than ground states, $T < 0K$; when all states are excited, $T = -\infty$.
As such, to cool a signal, we want a small positive temperature which corresponds to when there is more states in the ground state than the excited states. As in this case, to reach equilibrium, thermal photons will be absorbed by the system. This can be coupled to an MW signal to replace the cryogenically cooled termination.

Limitations

We hope this spin cold signal will be able to replace the cryogenic load operating at room temperature. These spin cold quantum systems, can absorb thermal photons as intended but can only be achieved within the linewidth of the exploited quantum transition.[44]
To be able to use this in experiments, the frequency of this mode needs to be adjusted or a d.c. magnetic field needs to be applied. These experiments are intrinsically narrowband, with the linewidth of this transition being a few MHz for pentacene, and a few hundred kHz for NV- diamond.[47] As such, when tuning a system to a cooling mode, one must take extra care to not mistakenly identify random spontaneous fading due to observing a narrowband, as mode cooling. This can be done by taking an average of independent measurements in quick succession.[44]

Chapter 3

Experimental

3.1 Samples

Yttrium Iron Garnet (YIG)

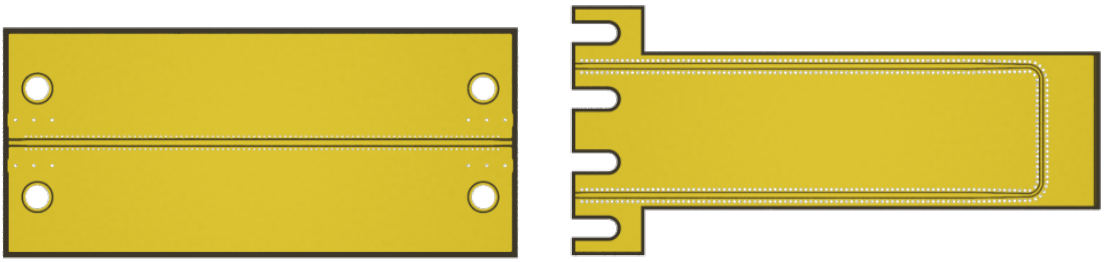
The YIG samples used in these experiments were provided by the University of Leeds, School of Physics and Astronomy. Two samples were deposited on a Gadolinium Gallium Garnet (GGG) substrate by RF magnetron sputtering with approximate thickness of 60 nm. The larger of the two samples was used to increase the contact area, measured to be 9.40×3.35 mm in length and width respectively.
The YIG samples were further tested using a Quantum Design MPMS3 Superconducting Quantum Interference Device (SQUID) to identify the substrate contribution to the magnetisation of the samples as well as the saturation magnetisation and coercivity.
The sample is mounted on the quartz sample holder which is then fixed to sample rods made of tapered carbon fibre tubes, with adapters on either end. The sample is vibrated at frequency ω by the measurement system about the centre of the detection coils, where the signal peaks as a function of sample position. The behaviour of the sample was measured between -10 to 10 Oe, which was expected to be enough to fully saturate the sample.

Permalloy (Py)

The Py sample used in these experiments was grown by Dr. Yoshii at Imperial College London's Blackett Laboratory. The sample was deposited on a SiO_2 substrate via evaporation deposition with an approximate thickness of 20 nm with an aluminium capping layer 10 nm thick. The length and width were measured, to approximately 11.25×9.15 mm respectively.
The Py sample provided was too large to fit within the SQUID sample holder, and as such was not tested due to risk of damaging the sample.

3.2 Coplanar Waveguide (CPW)

COMSOL is a multiphysics modelling software which uses finite element analysis to solve a number of partial differential equations (PDEs). COMSOL provides an RF (radio frequency) module optimised to simulate electromagnetic wave propagation. This module was used to simulate the scattering, S parameters of two waveguides using frequency domain analysis. All the materials used were the same ones as in the manufactured CPWs and are built-in materials within the COMSOL library.
The first waveguide design (Figure 3.1a) was designed according to the design constraints outlined in section 2.3 and consisted of a straight line trace with centre trace width, W =400 µm, and gap width, G =190 µm, resulting in a theoretical characteristic impedance of 62 Ω and had a straight taper on either connection to reduce insertion losses.
The second was design by Oliveira Schneider et al. [37] and provided in their GitHub under a general public license. This model was imported into COMSOL using the CAD import package. This U-path design, is exactly as used in [37], which at the sample position is defined by W_g =125 µm and G =125 µm with a resulting characteristic impedance of 50 Ω.



(a) Straight path CPW design. **(b)** U-shaped CPW design from [37]

Figure 3.1: KiCAD render of CPW designs tested.

Both waveguides were simulated to find the transmission and reflection coefficients, such that the power loss within the desired range of 0 to 15 GHz was acceptable. Which for the purposes of these experiments, was set to $S_{21} < -10$ dB.
The designs were transferred onto KiCAD, an open source software suit for Electronic Design Automation (EDA), for the purpose of manufacturing them through a third party manufacturer in PCBWay.
According to theoretical parameters as described in section 2.3 and further in appendix A, the waveguides were manufactured as two layer PCBs, using a 0.203 mm Rogers 4003c substrate; immersion tin for the surface finish, and 1 Oz copper thickness.

3.3 MuMax3 Simulations

MuMax3 is an open source, GPU-accelerated micromagnetic simulation program developed at the DyNaMat group at Ghent University.
The Py simulation was set such that the geometry would match that of the real sample, utilising periodic boundary conditions. The experiment was run with a field resolution of 50 MHz, and span static fields between 0 to 200 mT with 24 steps. Equivalent to a field resolution of 8.33 mT.
The YIG simulation was set to match the sample geometry utilising boundary conditions. The experiment ran with a field resolution of 50 MHz, but only had 10 steps between 0 to 200 mT.
The simulations were ran using the Google Colab feature provided by MuMax and Nvidia, as such the experiments were limited to around a 2 hour run time, and parameters had to be adjusted to fit within these constraints.

3.4 Broadband Ferromagnetic Resonance (FMR)

A VNA-FMR spectrometer setup was assembled for this work. The sample under test (SUT) is placed on top of a coplanar waveguide (CPW) with geometry described in section 3.2. The CPW is connected to ports 1 and 2 of a Copper Mountain vector network analyser M5180. The transmission coefficient, S_{21} is measured by taking 4001 points between 300 MHz and 17 GHz. Data is collected under no field for a background measurement. A personal computer is connected to both the VNA and the electromagnet power supply to control the experiment via a GPIB connection using SCPI (Standard Commands for Programmable Instruments). Measurements are taken at field strengths between 0 to 200 mT in ≈ 5 mT steps according to the calibration performed (See appendix B). The raw S_{21} parameter contains both the non-magnetic background and the magnetic response -the FMR signal. As such the background at 0 mT is subtracted from the collected data.

$$S_{21}^{\mathrm{FMR}} = S_{21} - S_{21}^{BG} \tag{3.1}$$

Where S_{21}^{FMR} is the FMR signal of interest, S_{21} is the raw signal, and S_{21}^{BG} is the background signal. This method may suffer from drifting sensitivity of the VNA but experimentally, it was sufficient to see a clear contrast in the signal without normalisation.[48,49] This experiment was carried out for SUT consisting of both the Py and the YIG samples. For each experiment, the measurements were performed 10 times and averaged to provide a clearer signal to noise ratio (SNR). The FMR signals were fitted to a symmetric + asymmetric Lorentzian fit (Appendix D) to obtain the Gilbert Damping parameter, gyromagnetic ratio, and linewidths in both frequency and field as described in section 2.1.

3.5 Inductive Magnon Noise Spectroscopy (iMNS)

iMNS experiments were carried out for both a hot source and cold source. Wherein for the hot source a Keysight 346B was used which outputs an ENF of 15 dB which is equivalent to a power of ≈−159 dBm. For the cold source a cryogenic 50 Ω termination resistor was dipped in liquid nitrogen at approx 77 K. A SignalHound Spectrum Analyser SA124B was used which has a detection floor of −147 dBm. As such three low noise amplifiers (LNAs) were connected in series between the CPW which is coupled to the SUT, and the SA which provide a total gain of 65 dB. The samples were placed directly onto the CPW and secured using Kapton tape. A simplified schematic of the set up is shown in figure 3.2. The CPW was mounted onto a 3D printed stand to ensure the sample was aligned with the centre of the electromagnet, and corresponding to the same point the field calibration was obtained.

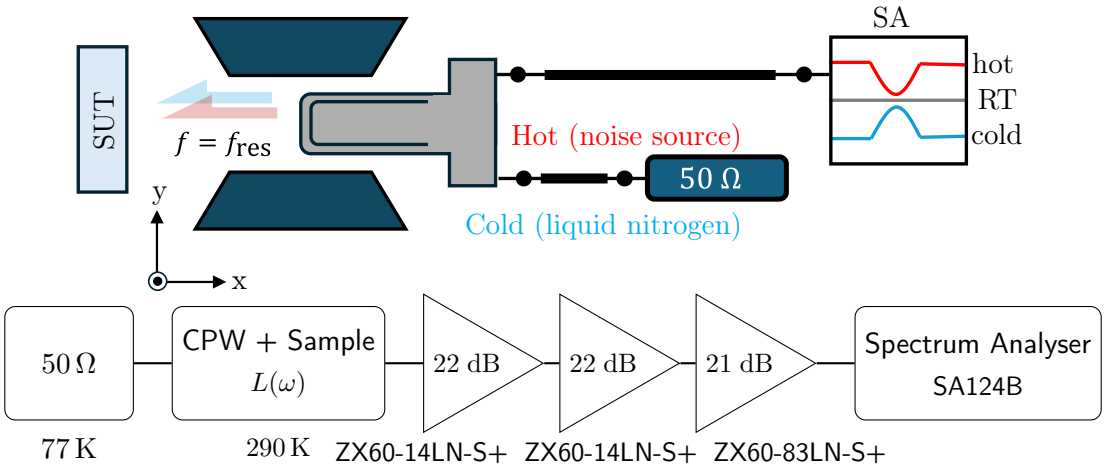


Figure 3.2: Simplified schematic of experimental set-up used for iMNS experiments. SUT is placed directly on CPW and experiences energy exchange both for cold (blue) signals or hot (red) signals. The expected signal shape is shown in the simplified SA. LNA chain not shown between CPW and SA is shown in system block diagram

To estimate the losses from the microwave system, the S_{21} parameters were taken for each individual component, as well as the overall system before and after the SUT.
The magnetic system should not influence the signal when at 0 field. As such background measurements were taken at 0 field. The experiments were at field strengths corresponding to the expected resonant field at 7 GHz ± 150 MHz from the data obtained in the VNA-FMR (Section 3.4) experiments.
The SA was set to 1 kHz radio bandwidth and 650 Hz video bandwidth (Appendix E) which for the SA used, has a limiting range of 50 MHz so 6 measurements were taken to achieve the 300 MHz bandwidth used.

Simulated iMNS

To confirm the expected results, the losses were used to simulate the results of the experiments using Python. Using the losses measured for previous VNA-FMR experiments (Section 3.4) the noise temperature of the system can be calculated using equation 3.2.

$$T_{\text{out}}(f) = T_{\text{source}} \cdot |S_{21}|^2 + T_{\text{env}} \cdot \left(1 - |S_{21}|^2\right) \tag{3.2}$$

Given the FMR experiments did not include the low noise amplifiers (LNAs) as part of the set up, the additional noise contribution can be calculated using a Friis chain as described in equation 3.3.

$$T_{\text{sys}}(f) = T_{\text{out}}(f) + T_{\text{LNA},1} + \frac{T_{\text{LNA},2}}{G_1} + \frac{T_{\text{LNA},3}}{G_1 G_2} \tag{3.3}$$

The temperature of each LNA can be calculated from their reference specification, where a noise figure is provided. This final system temperature, T_{sys} is used to calculate the spectral density using equation 2.9 to be able to compare with results found experimentally.

Chapter 4

Results and Discussion

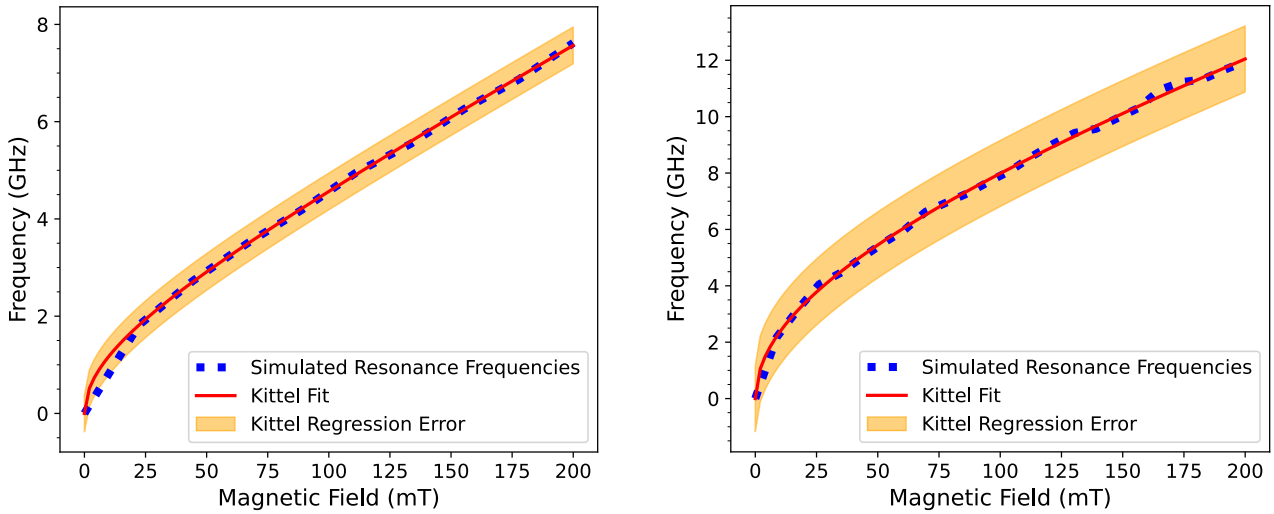
4.1 Samples

SQUID magnetometry was performed for the YIG sample with the applied field in-plane and out-plane. The out of plane experiments utilised field strengths too low to fully saturate the sample, so the anisotropy cannot be calculated.
After subtracting the paramagnetic contribution to the data, the coercivity and saturation magnetisation were measured to be 1.6 Oe and 194.7 kA m^{-1} respectively.

4.2 Simulations

MuMax3 Resonant Frequencies

The MuMax3 simulations were used to calculate theoretical values for the gyromagnetic ratio and saturation magnetisation for both samples as shown in figure 4.1. For the YIG simulation, the values of $\gamma/2\pi$ and M_{sat} were 27.9 ± 0.4 GHz/T and 132 ± 6 kA m^{-1}. While for the Py simulation, $\gamma/2\pi$ =29.7 ± 1.7 GHz/T and M_{sat} =496 ± 47 kA m^{-1}.



(a) Kittel fit of Mumax3 simulation of YIG sample

(b) Kittel fit of MuMax3 simulation of Py sample

Figure 4.1: MuMax3 simulation results, with Kittel fit extracted.

The resonant frequency as a function of field is fitted according to equation 2.1, using minimisation of the sum of squares of the residuals,[50] with the saturation magnetisation, and the gyromagnetic ratio as the fitting parameters.
The YIG sample needed a much longer run time due to the greater thickness of the sample. As such to comply with the allocated computation time, a rougher field sweep had to be performed. This may affect the accuracy of the calculated values. The Py simulation showed a much larger error than that of YIG, and it is

visible within the graph that the data points do not align as well with the fit. This could be attributed to the much smaller thickness of the film, resulting in magnon modes with a wavevector greater than 0, contributing to the final result. The overall fit is still adequate and is a good benchmark for the results obtained in later experiments.

COMSOL CPW S-Parameters

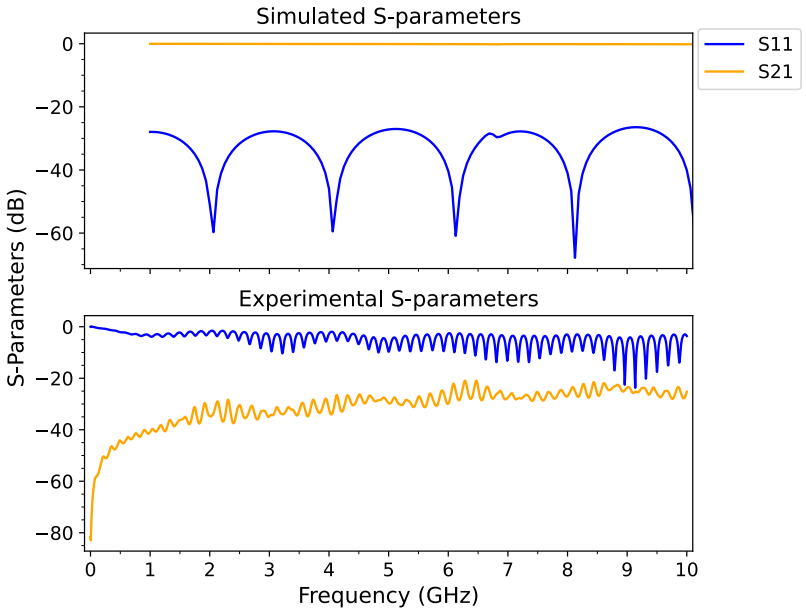


Figure 4.2: Scattering parameters of straight-path CPW. Comparing COMSOL simulation and experimental results. Experimental results show poor performance.

The straight path CPW design (Figure 3.1a) demonstrated very good S-parameters, with the S_{21} values being above $-5\,\mathrm{dB}$ as seen in figure 4.2. This is starkly contrasted by the experimental values wherein the S_{21} values were below $-20\,\mathrm{dB}$ for all frequencies.
This is postulated to be due to the conductor trace pad -the surface the SMA connector pin contacts with- was too far from the connector's pin given the position of the screw holes.

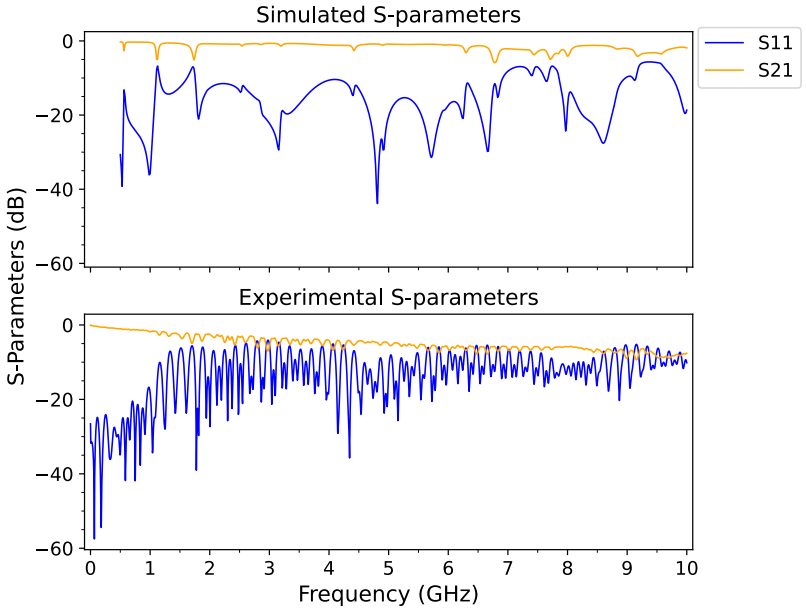


Figure 4.3: Scattering parameters of U-path CPW. Comparing COMSOL simulation and experimental results.

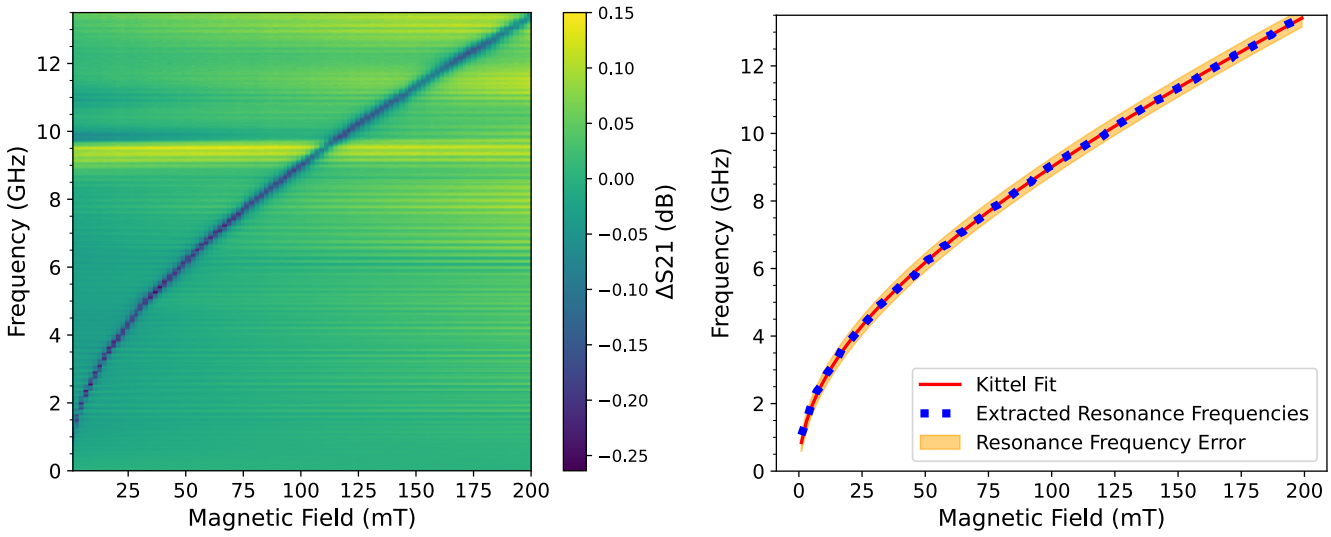
The U-shape CPW design (Figure 3.1b) displayed good simulation results with S_{21} values all above $-5.9\,\mathrm{dB}$

which agree with the experimental results wherein S_{21} was above $-9.3\,\mathrm{dB}$.
The straight-path CPW could likely be fixed by soldering the connector, but the U-shape design demonstrated very good S_{21} results which matched those of the simulations. Further, the U-shape CPW allows for a simpler experimental set up, with shorter coaxial cables, which will allow for less attenuation in iMNS experiments. As such, the U-shape CPW was used for all experiments.

4.3 VNA-FMR

Permalloy

VNA-FMR experiments for the Py sample, shown in figure 4.4 show a clear Kittel shape, for each individual experiment a Lorentzian was fitted and the parameters were used to extract a Kittel regression which was used to extract the gyromagnetic ratio, $\gamma/2\pi = 30.7 \pm 0.3\,\mathrm{GHz/T}$ and the saturation magnetization, $M_{\mathrm{eff}} = 605 \pm 12\,\mathrm{kA\,m^{-1}}$ by fitting them according to equation 2.1.
The full width half maxima of the Lorentzian fits were used as the frequency swept linewidth and used to calculate the field swept linewidths according to equation 2.5 using a linear regression. From these the value of alpha for the Py sample was measured to be $\alpha = 0.0084 \pm 0.0013$ and 0.0101 ± 0.0003 for frequency and field swept respectively, which agree with each other, and expected values from literature (citations). From the field swept fit, the inhomogeneous broadening, ΔH_0 was fitted to $0.00 \pm 0.19\,\mathrm{nT}$ which coincides with the assumptions made in section 2.1.



(a) ΔS_{21} heatmap for Py sample.

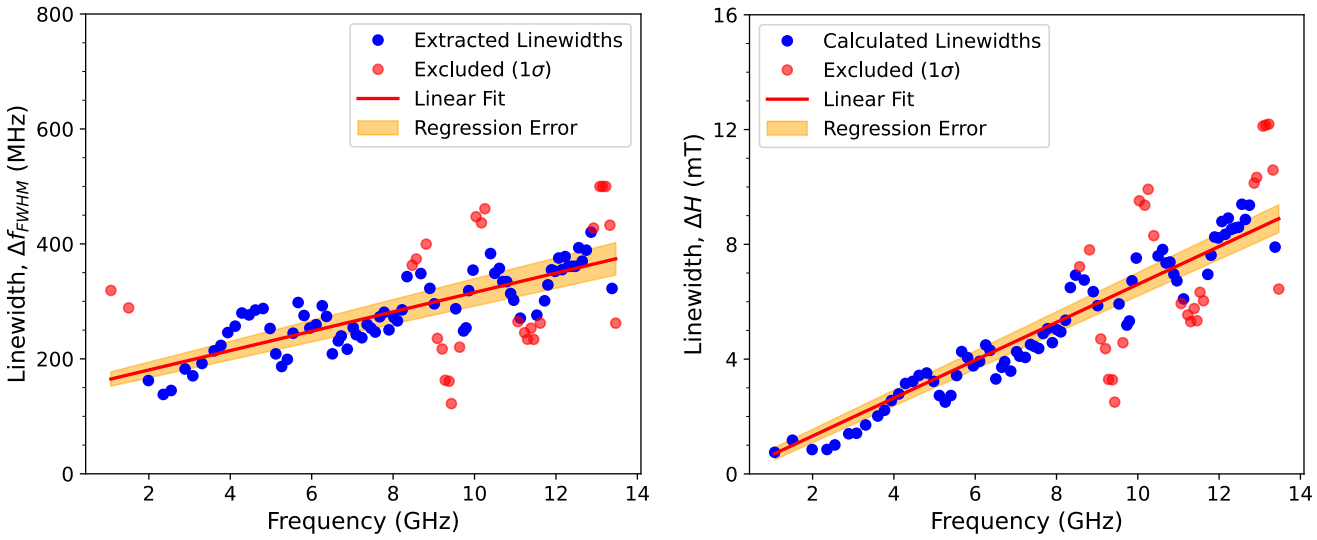
(b) Kittel fit from the extracted values from FMR.

Figure 4.4: VNA-FMR results for Py sample, showing a Kittel curve which is fitted in (b).

The resolution in figure 4.4a comes from the field resolution due to the power supply used to control the electromagnet. There is an apparent gain region at $\approx 9.5\,\mathrm{GHz}$, which would suggest the waveguide is emitting energy, which is unphysical given the nature of the experiment. It was postulated that this could be an unexpected consequence of the measured S_{21} parameters being mistakenly averaged in logarithmic units. This was investigated by simulating the random noise and averaging in linear and logarithmic units to observe any changes in extracted values. It was found that the numerical value of S_{21} were incorrect as a result of this, but the features of the Lorentzian fit remained consistent, and showed no gain regions. As such it was concluded that this would not affect the results.
This gain region was concluded to be due to a change in the losses between the background and the experiment data. This is likely as a result of handling the waveguides, which being less than 1 mm thick, were prone to bending.

YIG

The same VNA-FMR experiment was performed on the YIG sample, resulting in the Kittel relationship shown in figure 4.6. The extracted gyromagnetic ratio, and saturation magnetisation were $27.0 \pm 0.3\,\mathrm{GHz/T}$ and

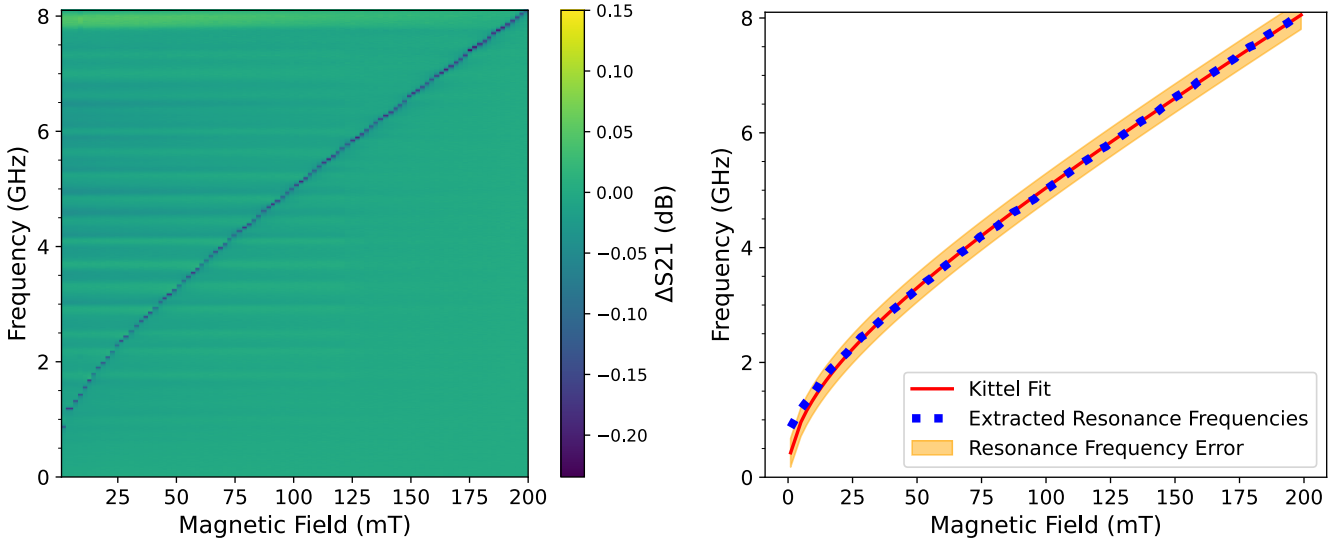


(a) Frequency swept linewidth, from Lorentzian fits.

(b) Field swept linewidth, from equation 2.5

Figure 4.5: Frequency swept and calculated field linewidths for Py. Data outside of a 1σ error were excluded from the fit.

$198 \pm 6\,\mathrm{kA\,m^{-1}}$ respectively. These values closely match those obtained in the SQUIDs experiment (Section 4.1) and in the MuMax3 micromagnetic simulations (Section 4.2).



(a) ΔS_{21} heatmap for YIG sample.

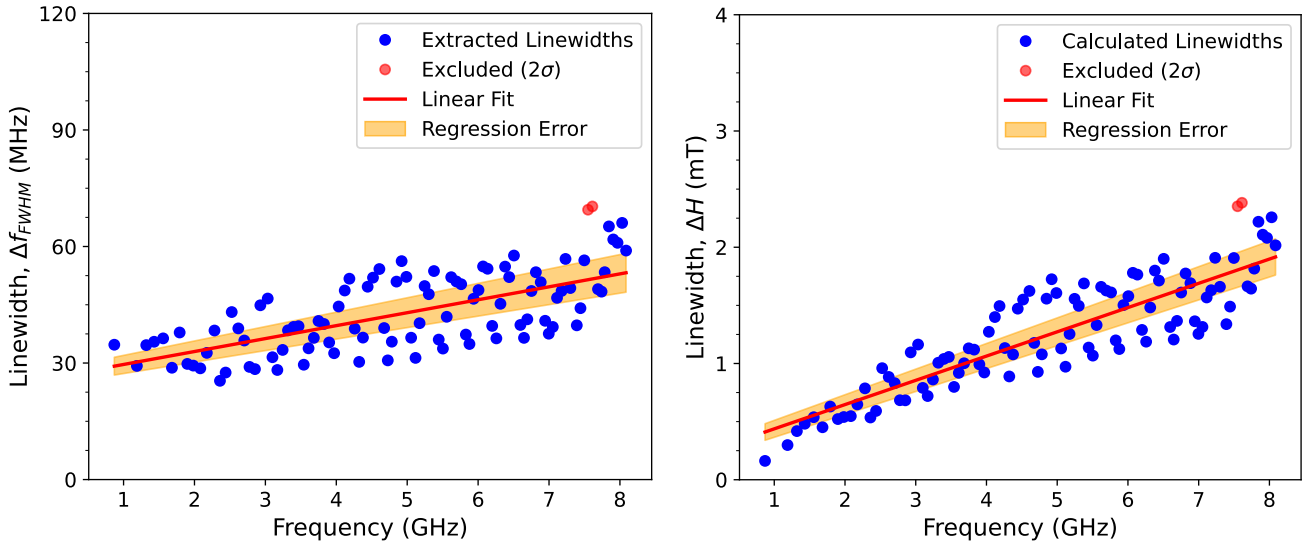
(b) Kittel fit from the extracted values from FMR.

Figure 4.6: VNA-FMR results for YIG sample, showing a Kittel curve which is fitted in (b).

Figure 4.6b shows the Kittel fit extracted from the Lorentzian fits of the FMR experiments. The error is the standard deviation obtained from the residuals of the regression used. The error is marginally larger for YIG than for Py, and the data appears to diverge from the fit at low frequencies. This could be due to the field resolution of the experiments, which as discussed previously, is approximately 5 mT, at these small fields, the resolution is not fine enough to accurately map the precise trend. Further it can be observed in figure 4.6a that at these lower values, the intensity of the resonance is much lower, as would be expected, which could lead to imperfect Lorentzian regressions, where the extracted peak centre was used as a fitting parameter for the Kittel regression.

It is immediately obvious that the YIG experiment had a much finer linewidth, shown by the thickness of the blue region in the heatmap in figure 4.6a. This is to be expected due to the known properties of YIG, and expectedly, the linewidths were shown to be several times smaller than those for Py. The linewidths were used to extract the alpha parameters, 0.0016 ± 0.0004 and 0.0028 ± 0.0002 for frequency and field swept

respectively. Which is nearly an order of magnitude smaller than that of Py. These results are consistent with literature for the sample geometry.[5] The homogeneous broadening was larger than expected with an extracted value of $0.23 \pm 0.06\,\text{mT}$. This could be attributed to the data points at low frequencies, which shift the regression up, and as discussed earlier, may be erroneous due to their small magnitude, which becomes more apparent in figure 4.7a as the first two values are outside of the 1σ range.



(a) Frequency swept linewidth, from FWHM of Lorentzian fits.

(b) Field swept linewidth, from equation 2.5.

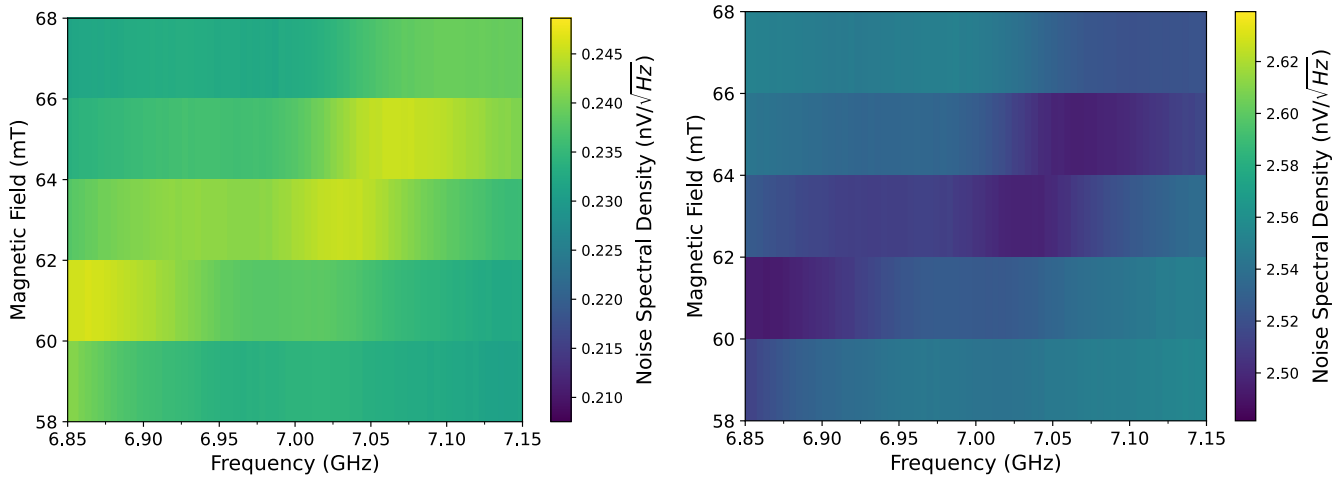
Figure 4.7: Frequency swept and calculated field linewidths for YIG. Data outside of a 2σ error were excluded from the fit. Shaded regions show the regression error from linear fit residuals

A gain region is seen in the YIG experiment at $\approx 8\,\text{GHz}$ similar to the Py experiment at $\approx 9.5\,\text{GHz}$, reinforcing this is a systematic error regarding the set up of the experiment and not influenced by the samples.

4.4 iMNS

Simulated Results

A simple simulation was run on Python using the S_{21} losses obtained via VNA-FMR experiments as outlined in section 3.5. The source temperatures used were 77 K and 9600 K for cold and hot sources respectively and 290 K was used as the environment temperature.



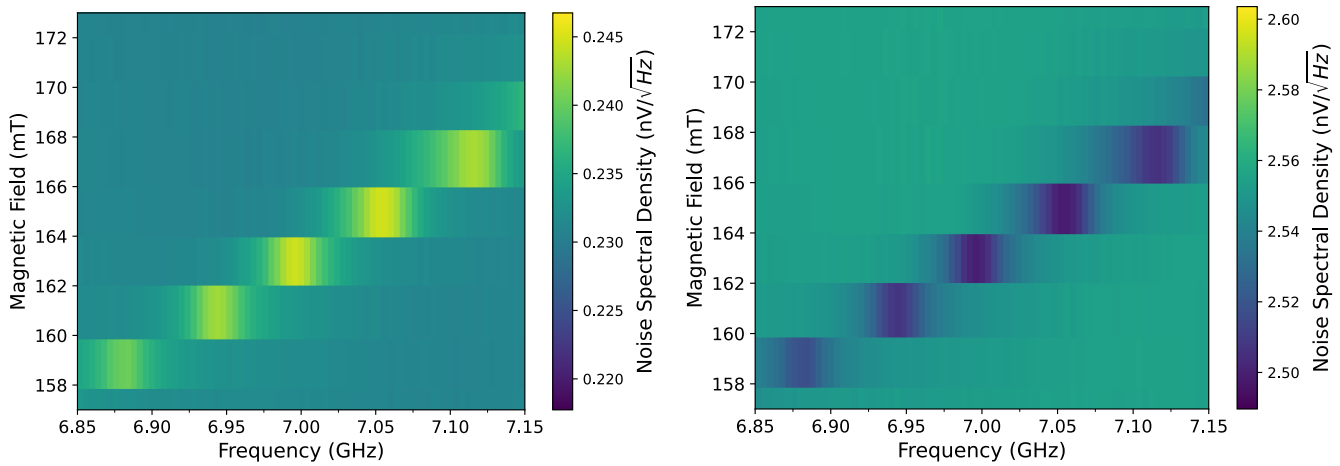
(a) Simulated heatmap for liquid nitrogen experiment.

(b) Simulated heatmap for hot noise source experiment.

Figure 4.8: Simulated iMNS noise spectral density results for Py

Figure 4.8 shows the simulated iMNS results for a Py film, between 6.85 to 7.15 GHz, the same frequency

range investigated in the iMNS experiments. There is clear contrast when $f = f_{\text{res}}$. As a consequence of linewidth, which for Py, $\Delta f_{\text{FWHM}} \approx 300\,\text{MHz}$ at 7 GHz, the resulting image is blurred, this is further emphasised by the low resolution of the simulation.
The simulation was performed using S_{21} parameters obtained from VNA-FMR (Section 4.3) which takes 4001 data points, the experimental data consists of 210 000 data points, so it should be expected that the resolution for the simulation will be worse. However, a discernable contrast is still clearly visible.



(a) Simulated heatmap for liquid nitrogen experiment.

(b) Simulated heatmap for hot noise source experiment.

Figure 4.9: Simulated iMNS noise spectral density results for YIG

As expected, the simulation for YIG, which has a narrower linewidth, $\Delta f_{\text{FWHM}} \approx 50\,\text{MHz}$ at 7 GHz, shows a more narrow feature at resonance.
The colour scales used in both experiments are scaled differently, to emphasise the contrast obtained for their respective experiments.

Experimental Results

Heatmaps using a fixed scale were plotted for both YIG (Figure 4.11) and Py (Figure 4.10), a maximum value for the spectral density was chosen to be $2.0\,\text{nV}/\sqrt{Hz}$ to maximise the contrast observed. This value is near the theoretical maximum value for the hot experiments ($\approx 2.56\,\text{nV}/\sqrt{Hz}$) which were not seen in any hot experiments, but values exceeding this were seen as anomalous spikes in cold and room-temperature experiments and show as bright yellow features.
Figures 4.10 and 4.11 show raw data, an attempt at background subtraction was made but were considered unsatisfactory, as no discernable contrast could be identified. This is discussed in greater detail in section 4.4 - 'Background Subtraction Methods'.

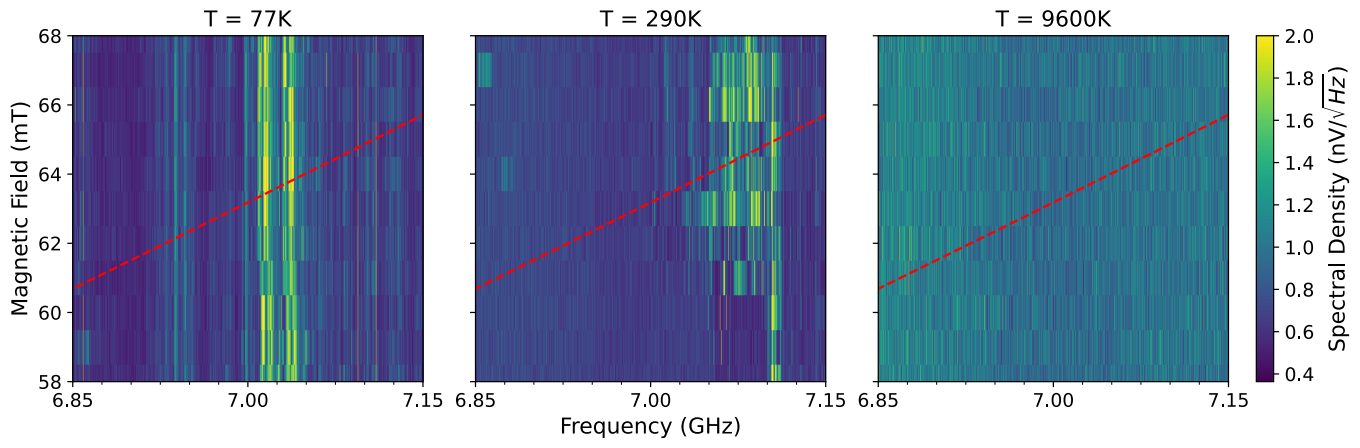


Figure 4.10: Heatmap plots for Py sample at $T = 77$ K, $T = 290$ K, and $T = 9600$ K. The same scale is used for all three plots. Red dotted lines indicate the location of expected contrast from VNA-FMR Kittel fit.

The experimental results did not show the resonant contrast expected from the simulations. This discrepancy is likely due to a combination of effects that compromised the absolute noise calibration, including changes in the CPW loss between calibration and measurement, hysteresis in the electromagnet, and artefacts introduced during background subtraction.
Large spikes are seen in the liquid nitrogen experiment, these spikes are not consistent with the S_{21} losses of the electric circuit and may be attributed to errors in the operation of the spectrum analyser as will be discussed further in section 4.4 - 'Analysis'.

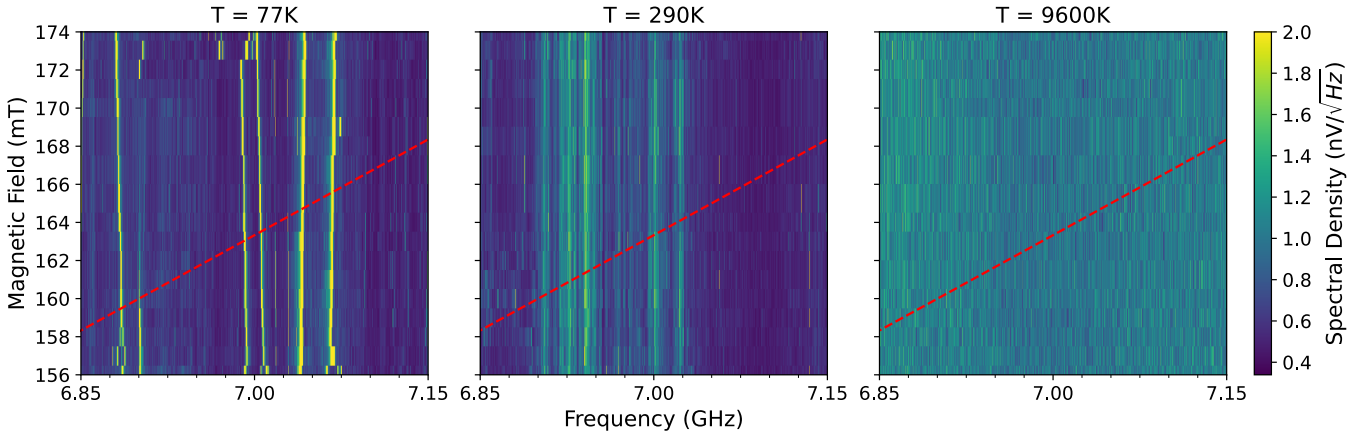


Figure 4.11: Heatmap plots for YIG sample at $T = 77$ K, $T = 290$ K, and $T = 9600$ K. The same scale is used for all three plots. Red dotted lines indicate the location of expected contrast from VNA-FMR Kittel fit.

The YIG heatmaps display the same anomalous spikes as Py. Wherein the cold experiments in figure 4.11 show multiple spikes which appear to be influenced by applied magnetic field. The room temperature data shows field independent spikes and regions cooler than expected.
Interestingly, the hot data, while still not displaying the expected contrast from simulations, displays a uniform spectral density with no large spikes or dips. This is consistent between both YIG and Py.

Analysis

The obtained data was further analysed to identify the source of the aforementioned anomalous artifacts. The data used for the heatmap in figure 4.10 at 59 mT is shown in figure 4.12. The top row is the data obtained from the SA, adjusted for gain from the LNA chain. The bottom row is the equivalent spectral density.
For each of the spectral density plots, the equivalent temperature spectral density was calculated using equations 2.9 and 3.3. The latter is labelled 'Friis' on the legend and was calculated using the average S_{21} parameters between 6.85 to 7.15 GHz to account for attenuation.
The room temperature displayed average noise level (DANL) is reasonable, but still displays spikes above 7 GHz which go beyond that which is physical. The cold DANL is also reasonable compared to the level expected accounting for losses (the 'Friis 77K' line on figure 4.12) but it displays an oscillation on its amplitude output which could be an effect of the SA. However, these oscillations greatly exceed the magnitude of error reported by SignalHound of ±2.5 dB.
The same analysis was performed for the data obtained for YIG at 156 mT. The data follows a similar trend. The DANL for room temperature is reasonable, but there is a lot of unexpected noise which raises this floor above the expected level between 6.9 to 7.0 GHz and lowers it above this. As a result the spectral densities for the room temperature and cold experiments are at similar levels.
The yellow spikes seen in figure 4.11 can be easily spotted in figure 4.13, both in the spectral density and the amplitude plots. Showing they occur independently of applied field strength.
For both Py and YIG, the hot data shows a clear DANL and the amplitudes measured show no anomalous spikes or drops. However, they are both significantly lower than would be expected according to the ENR given by Keysight. Which is at approximately the same level for both with and without a Friis chain. This may imply the losses in the system are higher than hinted at by the S_{21} parameters obtained in VNA-FMR experiments. The most probable source for a difference in loss parameters (besides the LNA chain which has been taken into account) would be resulting damage to the waveguide used for the experiments. During

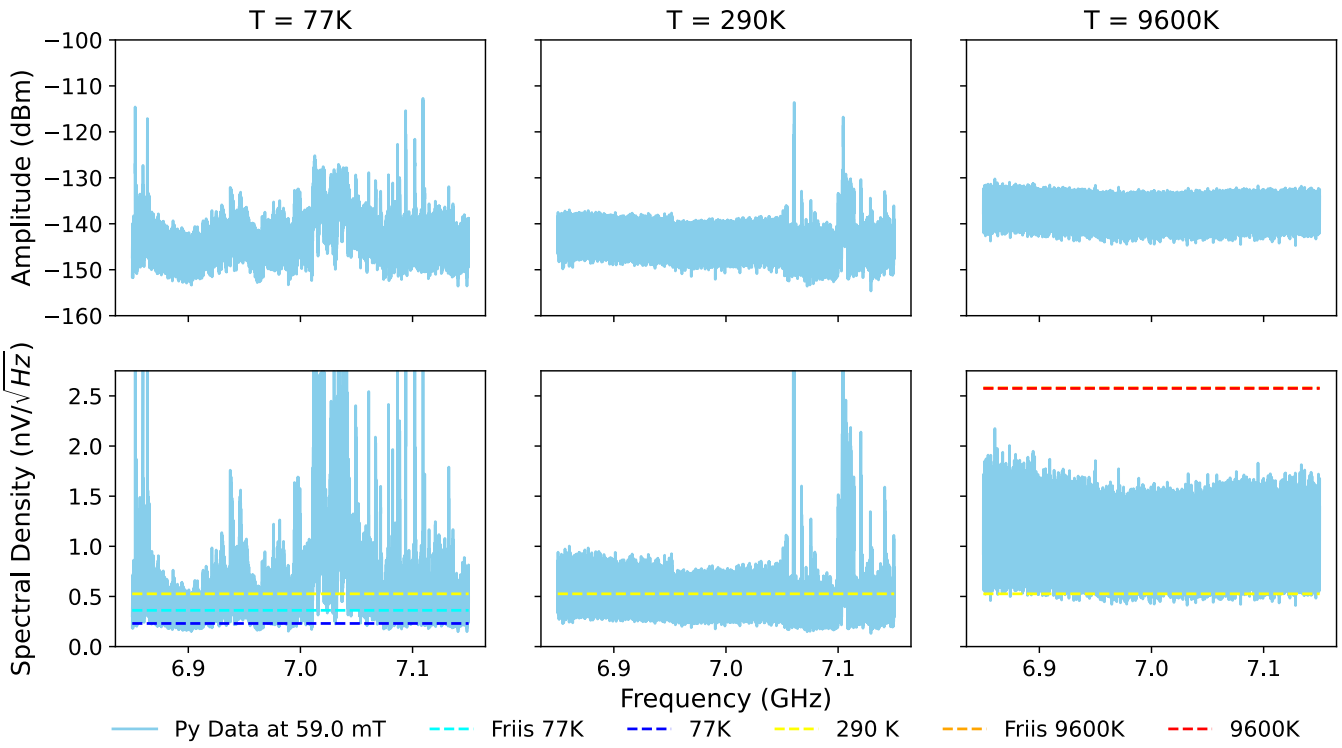


Figure 4.12: Experimental measurements (top row) and corresponding spectral densities (bottom row) for Py sample using noise source temperatures of $T = 77$ K, $T = 290$ K, $T = 9600$ K. Noise temperatures with and without accounting for losses are shown on spectral densities. Calculated using equation 3.3 and 2.9 respectively.

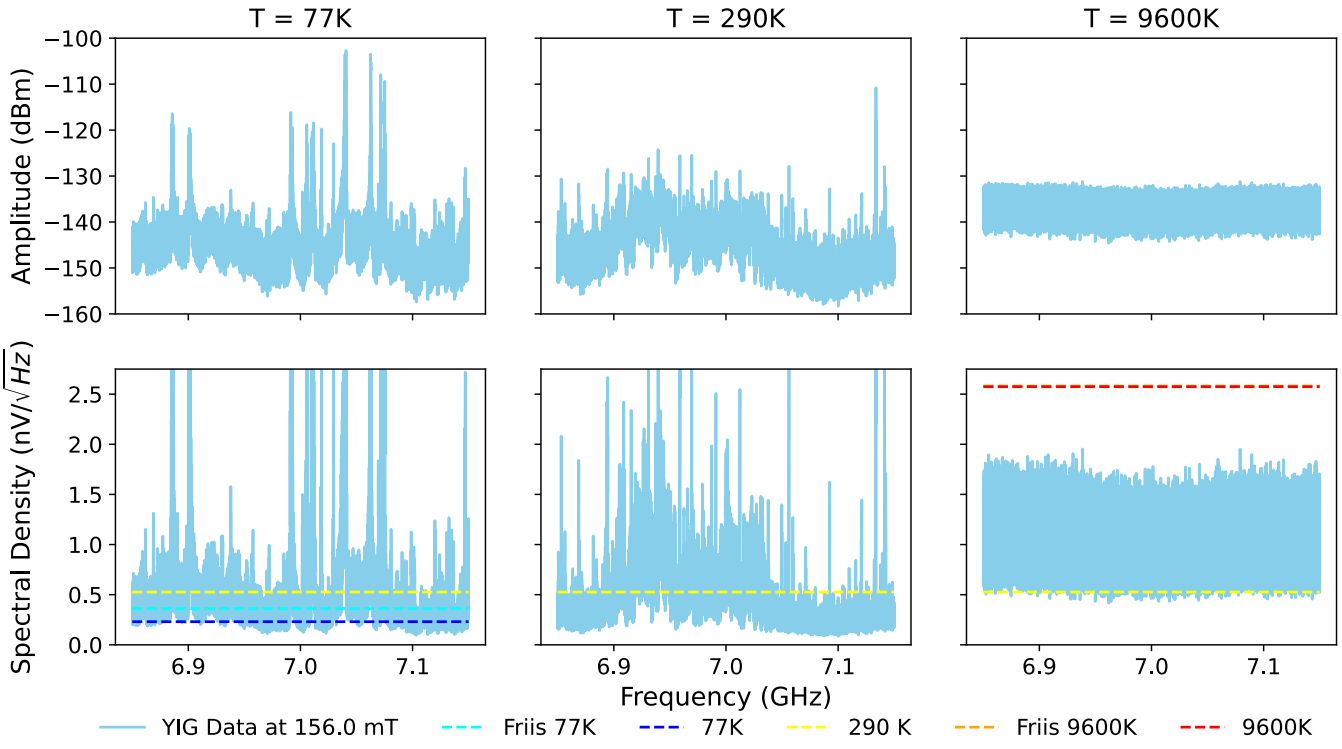


Figure 4.13: Experimental measurements (top row) and corresponding spectral densities (bottom row) for YIG sample using noise source temperatures of $T = 77$ K, $T = 290$ K, $T = 9600$ K. Noise temperatures with and without accounting for losses are shown on spectral densities. Calculated using equation 3.3 and 2.9 respectively.

experimental work, it was noted that the waveguide due to its thickness would bend easily potentially damaging the waveguide. Great care was taken to avoid this, including the making of a 3D printed stand for

the waveguide, but any handling would result in small damage.
This explanation is consistent with the timings of the experiments - cold measurements were carried out after hot measurements.

Data Acquisition

From visual inspection of the data, periodic patterns could be recognised. This may be an artifact of the manner in which the Python script operated.
The code which controlled the experiments operated by sampling all fields while at each given span of the spectrum analyser (50 MHz span) which could affect the obtained results. This could manifest as hysteresis in the electromagnet or by affecting the magnetic history of the samples.

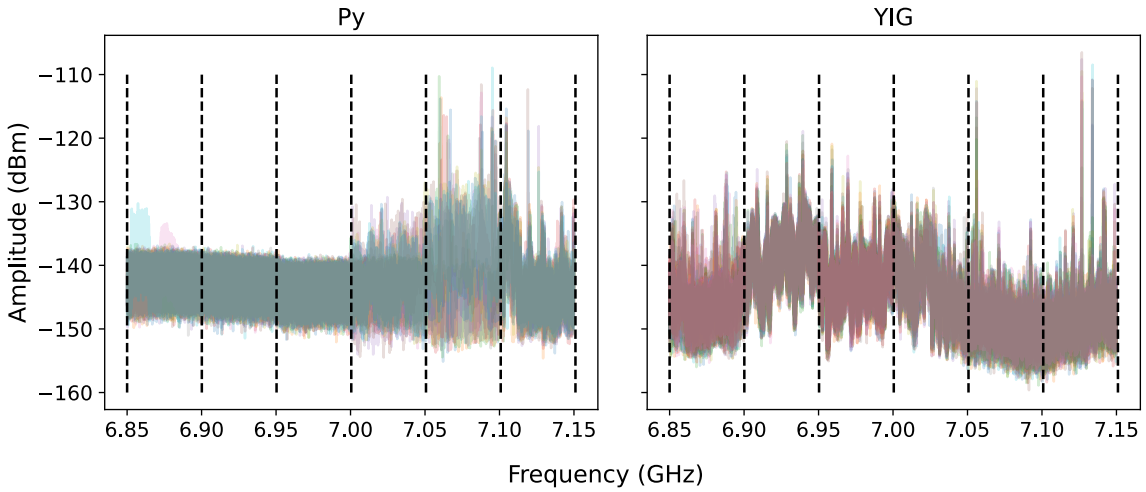


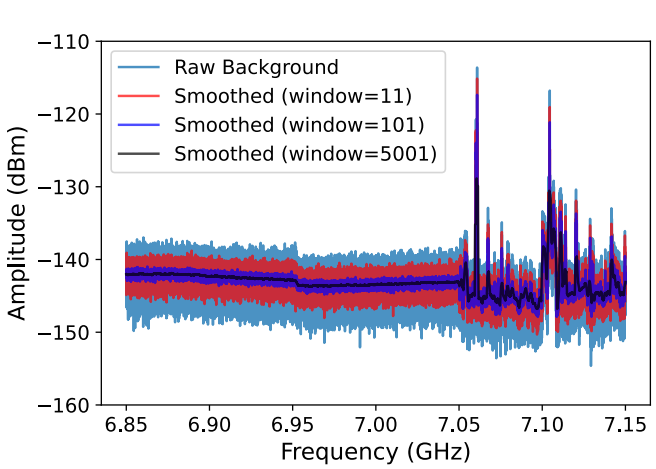
Figure 4.14: Experimental amplitudes for room temperature experiments for both Py (left) and YIG (right). Results for each field strength plotted with low opacity. Dotted lines indicate the span for each SA instance with a 50 MHz span. Field independant noise can be observed at each slice.

Figure 4.14 shows where each instance of the SA was initialised, with all room temperature data overlaid with low opacity for Py and YIG.
Trends can be identified within each of the segments. However, these are not the same across segments which one would expect to see if it were a systematic artifact of the SA. Further data correction methods could be implemented by treating each instance individually. Due to time limitations these were not carried out.
The SignalHound User Manual states to keep the SA below 50 °C, it is unlikely the device ever exceeded this limit but it was in operation nearly continuously for 7 hours.
Another consequence of the sequence the experiments were performed in, is in the operation of the electromagnet. A GMW 3480 Dipole Electromagnet was operated using a Sorensen DLM80-37E power supply through its GPIB port using SCPI. The power supply has a current accuracy of 0.4 A, which was not capable of reliably changing the current output by intervals smaller than equivalent to 2 mT according to the calibration that was performed (Appendix B). (The current accuracy refers to how close the actual current is compared to the current requested.) As a result data obtained for what a given expected field strength may have been off by up to 4 mT, although these discrepancies were known from the 'read current' feature of the power supply. As such when the fields drifted between frequency intervals, the amplitude values were interpolated from the nearest data points.
As mentioned earlier, electromagnets are susceptible to hysteresis. The effect this may have on the applied field could not be known from the set up used in experiments. Raising the risk of the output field not corresponding to the calibration, an experimental set up which uses as teslameter as part of the equipment could mitigate this, but greatly increases the complexity of the set up. Further, cycling through the fields multiple times in one experiment could affect the results by affecting the magnetic history of the samples. However, this is unlikely within the conditions of the experiments performed. Because magnetic history tends to occur below saturation which for the purposes of these experiments, the samples were fully saturated and displayed negligible inhomogeneous broadening (Section 4.3).

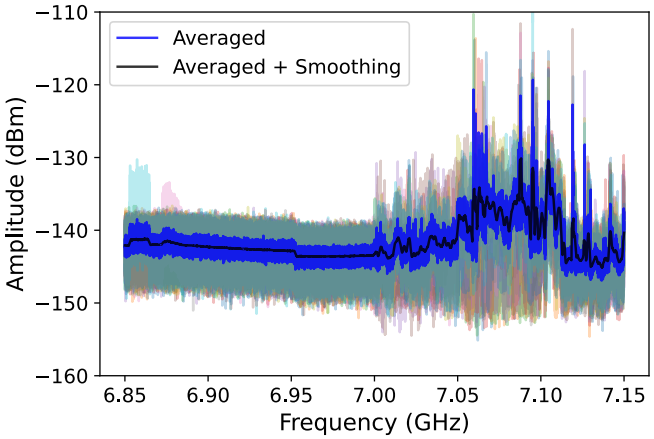
Lastly, the experiments may have suffered from a rough field resolution, depending on fewer measurements to observe contrast. YIG had an estimated field-swept linewidth of 2 mT and 4 mT for Py. Which is smaller than the field resolution used in the experiments due to the limitations in controlling the power supply and therefore to control the electromagnet. However, as was shown in the simulated results in section 4.4, contrast is able to be seen even with these restrictions, and is not a limiting factor.

Background Subtraction Methods

To be able to identify contrast within obtained data, a background subtraction is performed to separate systematic errors such as drifting. Background data with no applied magnetic field was attempted to be obtained, but due to an issue in the data acquisition code all data above 50 MHz were null. As a consequence of this, multiple background subtraction methods were attempted and multiple different data sets were used with no observable difference between them.



(a) Effect of smoothing for different window sizes. Smoothing consists of a rolling average.



(b) Low opacity data for each field overlaid. Average amplitude across all fields shown in blue. Averaged and smoothing shown in black, using a smoothing window = 5001.

Figure 4.15: Data processing methods used and their effect on the data. All processes calculated in linear units before correcting to dBm. Py data at room temperature used for demonstration.

Figure 4.15a shows raw data for Py at room temperature and at 59 mT with increasing amounts of smoothing overlaid. The smoothing is a rolling average with a window size corresponding to the number of data points averaged. With increasing amounts of smoothing, the spread of the data decreases. The displayed average noise level (DANL) reduces quickly but features with high contrast remain even with very large window sizes. Thereby smoothing data on its own is not sufficient processing for background subtraction, as spurious responses will remain and show contrast in the final results.
Figure 4.15b shows the data for all fields measured for room temperature experiment overlaid at low opacities, above 7 GHz there is clear discrepancies between each measurement. (note that the experiments were at room temperature, and as such the magnetic field should not have an effect on the microwave system). Averaging across all fields was done to remove some of these peaks and leave only the trends. Further in the figure the averaged data is smoothened using a window size of 5001 points, which gives a more consistent floor.
It is worth highlighting that to keep the meaningful numerical values after data processing, the background must be subtracted and then the data must be returned to its DANL by adding an offset. In the experiments carried out by Siegl et al. [1], they performed a polynomial regression on their data and used the first coefficient as their offset.
A similar method was attempted but fitting a regression to the data was not achieved, as such the average and smoothed background was used, and the offset was set to the average level which may introduce error in the numerical value, but doesn't hide any contrast.

Chapter 5

Conclusions

5.1 Outcomes

Two CPWs were manufactured, a straight path one, and a U shaped one. The scattering parameters were measured experimentally and simulated using COMSOL. The second, U-shaped design was chosen for the subsequent experiments for its suitable transmission coefficient, S_{21} loss of $-9.3\,\mathrm{dB}$ and simple design which could be mounted on a simple 3D printed stand for positioning.
MuMax3 micromagnetic simulations were carried out for the sample geometries of the Py and YIG samples to be tested. The resulting relaxations were fitted onto a Kittel fit with the gyromagnetic ratio and saturation magnetisation as the regression parameters.
VNA-FMR experiments were used as a backbone for subsequent iMNS experiments. In these the gyromagnetic ratio and saturation magnetisation were also fitted to a Kittle equation. Resulting in $\gamma/2\pi = 30.7 \pm 0.3\,\mathrm{GHz/T}$ and $605 \pm 12\,\mathrm{kA\,m^{-1}}$ for Py, which is closer to the literature values than for the micromagnetic simulation. And $\gamma/2\pi = 27.0 \pm 0.3\,\mathrm{GHz/T}$ and $198 \pm 6\,\mathrm{kA\,m^{-1}}$ for YIG. Which agree with prior SQUID experiment and largely agree with the micromagnetic results as well.
More crucially for use in spintronic technologies, the Gilbert damping parameters were fitted from the linewidth of these experiments. The YIG sample was found to have a Gilbert damping parameter, α of 0.0021 ± 0.0008; for the Py sample it was found to be 0.009 ± 0.002, both of which are consistent with literature. Low inhomogeneous broadening was observed, which is consistent with thin film conditions.
iMNS simulations were performed taking into account the experimental conditions. These provided clear contrast in the output heatmaps. Both for YIG and Py simulations, the spectral densities were of expected orders of magnitude ($\approx 0.23\,\mathrm{nV}/\sqrt{Hz}$ for cold experiments, and $\approx 2.5\,\mathrm{nV}/\sqrt{Hz}$ for hot experiments). These results confirm the feasibility for iMNS as a tool for analysing equilibrium magnetisation dynamics.
The experimental iMNS results were inconclusive for both samples showing no clear contrast. Multiple background subtraction methods were attempted but showed no discernible improvements in final heatmaps produced. It was postulated that resulting damage to the waveguide had affected the transmission coefficient between measurement of data and backgrounds. Further, using a spectrum analyser which allows for an increased span of 150 MHz would allow for more consistent data acquisition, and simpler background subtraction.

5.2 Further Work

IT was noted that it might be possible to extract meaningful contrast from the existing data with more rigorous data analysis. However the following modifications were discerned to the experimental method would allow for simpler data analysis resulting in resonant contrast. Using a larger bandwidth spectrum analyser to reduce ambiguity for background subtraction methods; perform all experiments for one sample at a time, ensuring impedance is the same across experiments; and ensure the experiment loops across all frequencies before fields to be measured.
If this work were to be attempted again, it would be advisable to use a thicker CPW to ensure mechanical stability during and between the experiments. The thickness of the CPWs used led to them bending and twisting which is believed to be one of the leading reasons for spikes in the data, due to impedance differences leading to reflections being different across experiments.

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Appendix A

CPW Design

A.1 Waveguide Losses

A CPW has losses attributed to two primary factors: dielectric losses, and conductor losses. These are defined by the materials used and their corresponding thicknesses. The rest of the radiative losses can be controlled through the geometry by minimising resonance conditions and reflections.

Via Placement

A via is a drilled hole that connects two or more metal layers through the dielectric of a printed circuit board (PCB). Via fences are common practice in MW electronics as they help reduce resonant modes at higher frequencies.
At a resonant frequency, the electromagnetic energy couples to the side grounds, thus reducing transmission along the trace.[51] The resonant frequencies can be calculated from the dimensions of the grounded sides, assuming they are symmetric, by approximating the side grounds as rectangular patch antenna. Because the dielectric thickness (H) is electrically very small $H \ll \lambda$, the resonant frequencies are given by:[51,52]

$$(f_r)_{mn0} = \frac{c}{2\sqrt{\varepsilon_r}} \sqrt{\left(\frac{m}{L}\right)^2 + \left(\frac{n}{\mathrm{SGW}}\right)^2} \tag{A.1}$$

Where the mode numbers, m, n, p are positive integers where at least one must be non-zero and $p = 0$ for the given thin geometry.
As a result, resonances are pushed to higher frequencies as the SGW is reduced. The via fence acts to reduce the electrical SGW with an effective value, $\mathrm{eSGW} = \mathrm{VL} - \mathrm{G} - \mathrm{VR} - \frac{\mathrm{S}}{2}$. Indicating the signal conductor should be as close to the via fences as possible.
Lastly, It is well established that the via spacing should remain below the $\lambda/4$ criterion, corresponding to the cutoff frequency $f_c = c_0/(4d_v\sqrt{\varepsilon_{eff}})$ with the speed of light in vacuum c_0, the via pitch d_v, and the effective dielectric constant ε_{eff}.[37,51]

Tapers

A taper is used to change the geometry of a transmission line to reduce reflections. A smooth transition between two widths will result in fewer reflections that a sharp one. These can be used to connect two lines of the same impedance but different widths, or lines with different width and impedances.
There are linear, polynomial, exponential, and Klopfenstein tapers.[53] Which all aim to do the same task with different levels of allowed reflections, Klopfenstein typically being capable of reaching well below $-20\,\mathrm{dB}$

Appendix B

Electromagnet Calibration

A Group 3 Teslameter was used to calibrate the field strengths at a given current power supplied by the power supply. 10 measurements were taken at each applied current and their standard error of the mean is

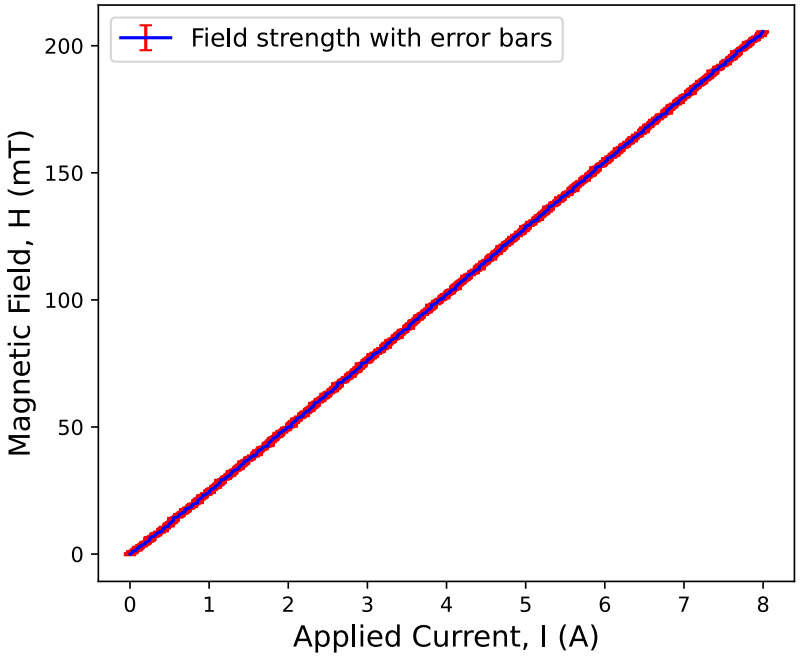


Figure B.1: Calibration curve of measured field at centre of coils for applied current by power supply.

plotted. Error bars are too small to be seen in the plot.
Every sequence of code being run would check for a calibration file before operating and would use the most recent one.

Appendix C

CPW Stand + Set-ups

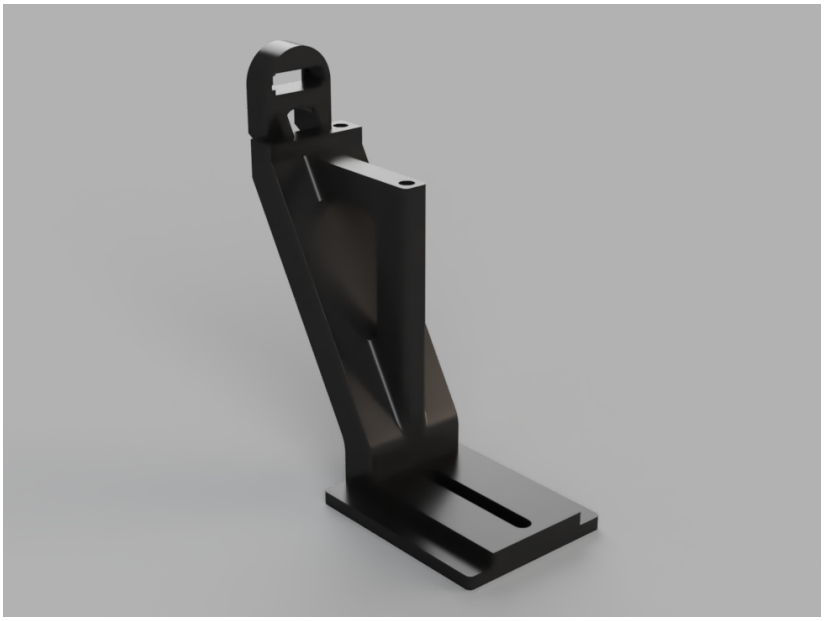


Figure C.1: Fusion 360 render of 3D printed stand used in FMR and iMNS experiments.

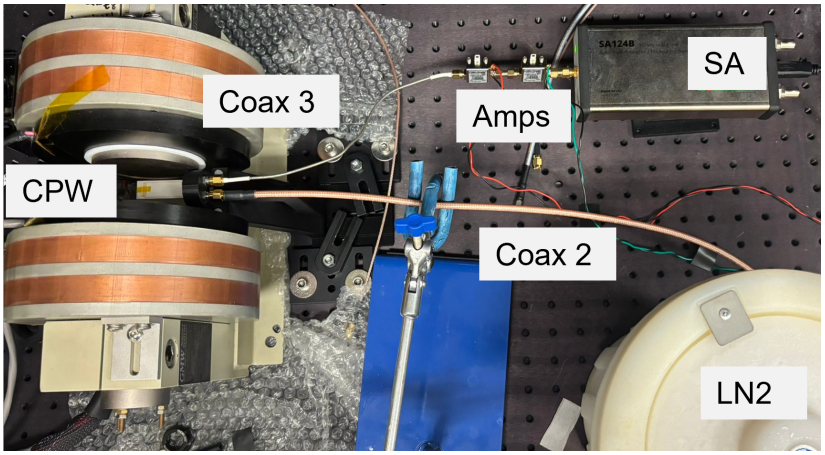


Figure C.2: Picture of experimental set up for iMNS cold experiments. Third LNA is missing.

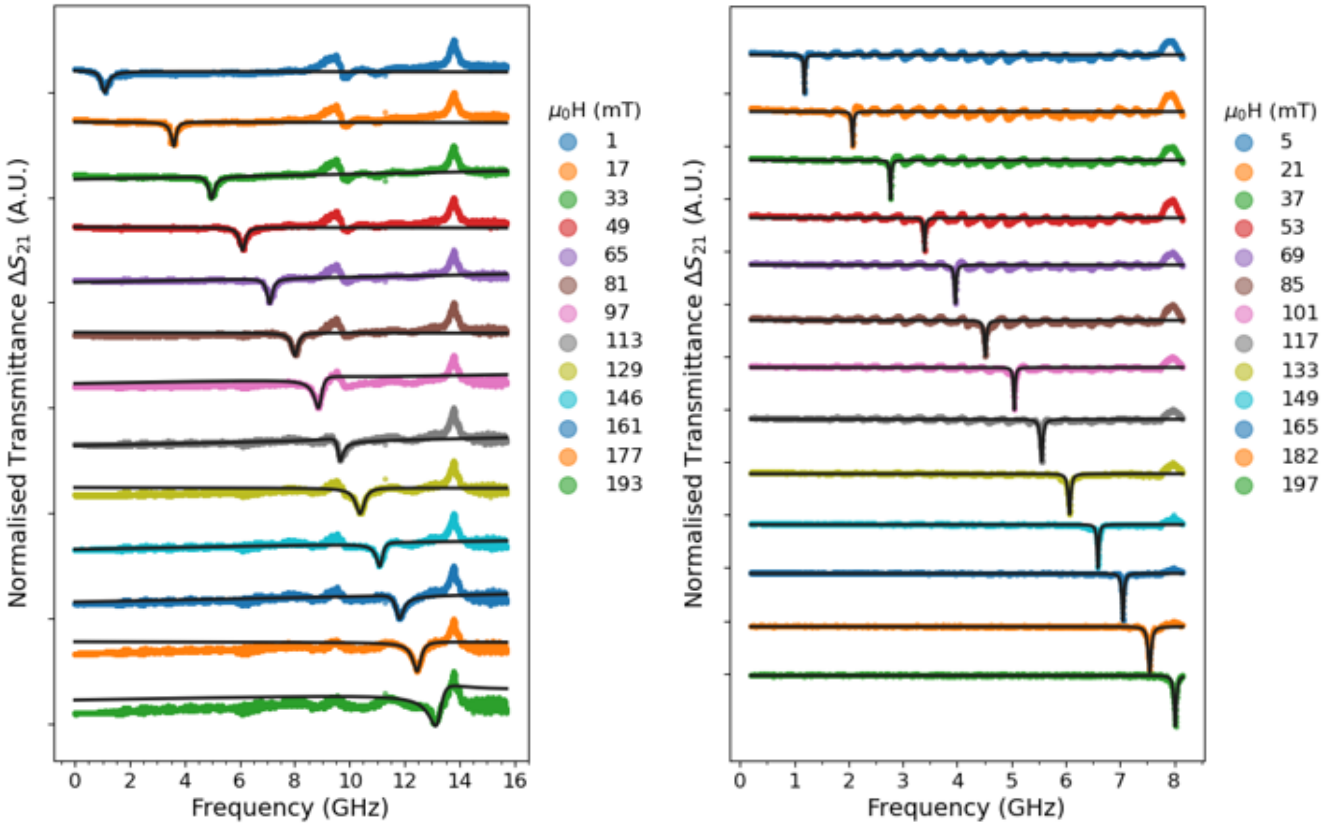
Appendix D

FMR Lorentzian Fitting

Near resonance, the lineshape is given by an asymmetric Lorentz line shape of the form:

$$A_0^{\mathrm{SYM}} \frac{\Delta f_{\mathrm{FWHM}}^2}{(f - f_{\mathrm{centre}})^2 + \Delta f_{\mathrm{FWHM}}^2} + A_0^{\mathrm{ANT}} \frac{\Delta f_{\mathrm{FWHM}}(f - f_{\mathrm{centre}})}{(f - f_{\mathrm{centre}})^2 + \Delta f_{\mathrm{FWHM}}^2} \tag{D.1}$$

Where A_0 is the amplitude of the symmetric and antisymmetric Lorentzian contributions respective to their superscripts. f_{centre} is the positioning of the Lorentzian, and Δf_{FWHM} is the full width half maxima of the Lorentzian. For the purpose of these experiments, at each magnetic field strength the plots were fitted to equation D.1 + an offset + gradient. Where amplitudes, FWHM, center, offset, and gradient were all fitting parameters.



(a) **(b)**

Figure D.1: Map of Lorentz fittings used at different magnetic fields for Py VNA experiments (a) and for YIG experiments (b)

Appendix E

Operating of Spectrum Analyser

A spectrum analyser operates by passing a signal over a filter which only allows certain frequencies through and their power is measured and displayed. This filter is moved across the span set for the device. Further processing may be done to smoothen the signal.

E.1 Radio Bandwidth

Noise is a random signal whose power is distributed over its usable bandwidth. This results in the power measured being proportional to the bandwidth. This must be considered because as a system's bandwidth increases, more information can be processed, but will also produce more noise.
Importantly, sweep time increases with a better (or lower) bandwidth. One can write a simplified equation for sweep time as:

$$\text{Sweep Time} = \frac{k_{\text{total}}.\text{SPAN}}{\text{RBW}^2} \tag{E.1}$$

Where k is the number of frequency selections under span. Which is to say that the sweep time is proportional to span and inversely proportional to the square of RBW.

E.2 Video Bandwidth

Video bandwidth is a time-domain low-pass filter, similar to an average. The main effect of VBW is smoothing the trance, and reducing noise.
It is important to note that VBW is a post process, so it is applied after the data is collected, but before being displayed.